



# **034IN - FONDAMENTI DI AUTOMATICA - FUNDAMENTALS OF AUTOMATIC CONTROL**

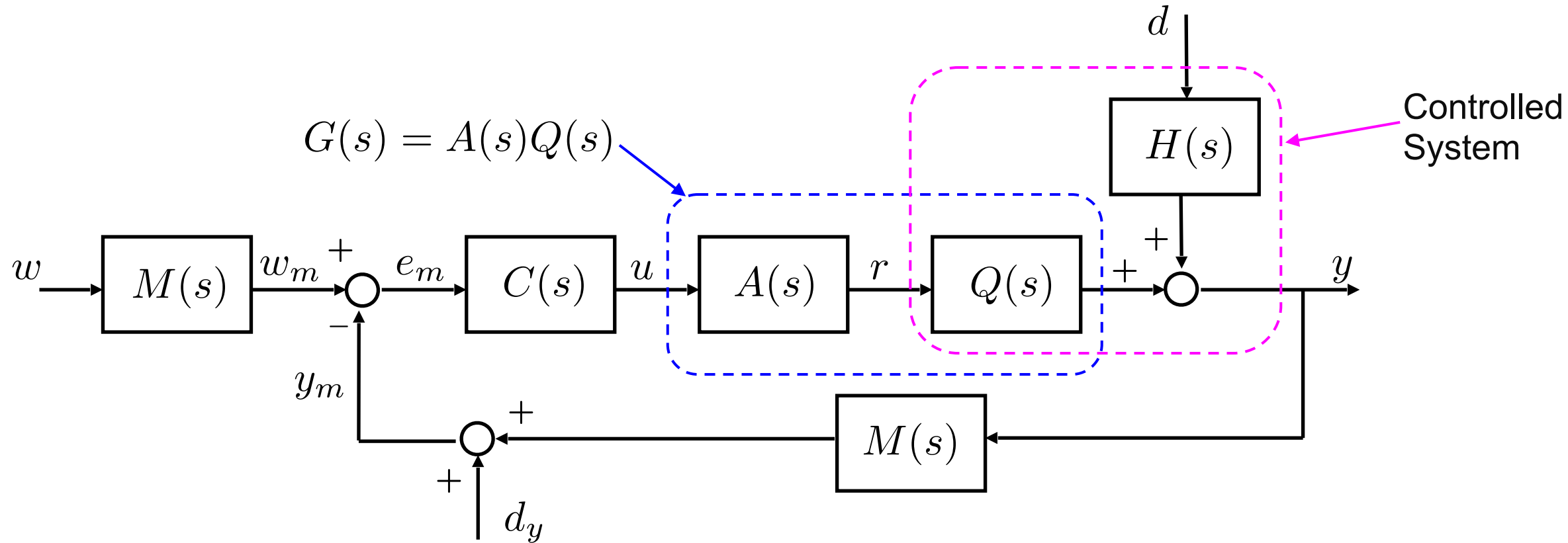
**A.Y. 2024-2025**

**Part X: Design of Feedback Control Systems**

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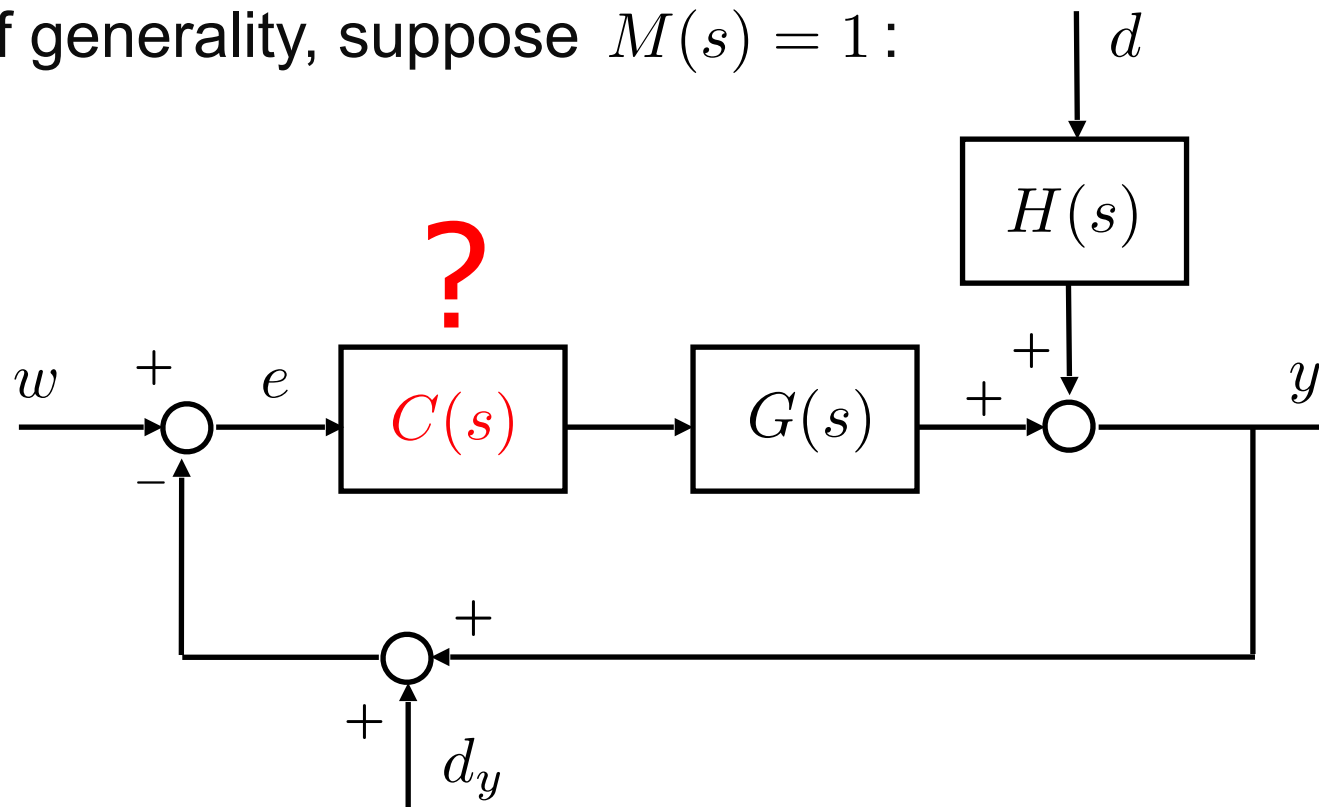
Department of Engineering and Architecture

Recall from Part 9, slide 10 (adding the output disturbance  $d_y$ ):



# Design of the Controller

Without loss of generality, suppose  $M(s) = 1$ :



- **Controller Design:** Determine the transfer function  $C(s)$  of the controller such that the **closed-loop requirements are satisfied**
- **Very different** from the analysis dealt with in [Part 9](#): now the open-loop transfer function  $L(s) = C(s)G(s)$  is **unknown** because  $C(s)$  is **unknown**!

# Closed-Loop Qualitative Requirements on Bode Diagrams

- Closed-loop asymptotic stability  $\longrightarrow \left\{ \begin{array}{l} \mu > 0 \\ \varphi_m > 0 \end{array} \right.$
- Static precision  $\longrightarrow \left\{ \begin{array}{l} g > 0 \text{ (pole(s) in 0)} \\ \text{and/or} \\ \mu \text{ "large enough"} \end{array} \right.$
- Dynamic precision
  - speed of the closed-loop response  $\longrightarrow \omega_c \text{ "large enough"}$
  - closed-loop damping ratio  $\longrightarrow \varphi_m \text{ "large enough"}$



- Disturbance rejection on direct path  $\longrightarrow \left\{ \begin{array}{l} \omega_c \text{ "large enough"} \\ |L(j\omega)|_{\text{dB}}, \omega < \omega_c \text{ "large enough"} \end{array} \right.$
- Disturbance rejection on feedback path  $\longrightarrow \left\{ \begin{array}{l} \omega_c \text{ "not too large"} \\ |L(j\omega)|_{\text{dB}}, \omega < \omega_c \text{ "small"} \end{array} \right.$
- Robust stability  $\longrightarrow \left\{ \begin{array}{l} \varphi_m \\ K_m \end{array} \right. \text{ "large enough"}$

Starting from closed-loop requirements, precise specifications on Bode diagrams are devised

## Example

- Static Specifications

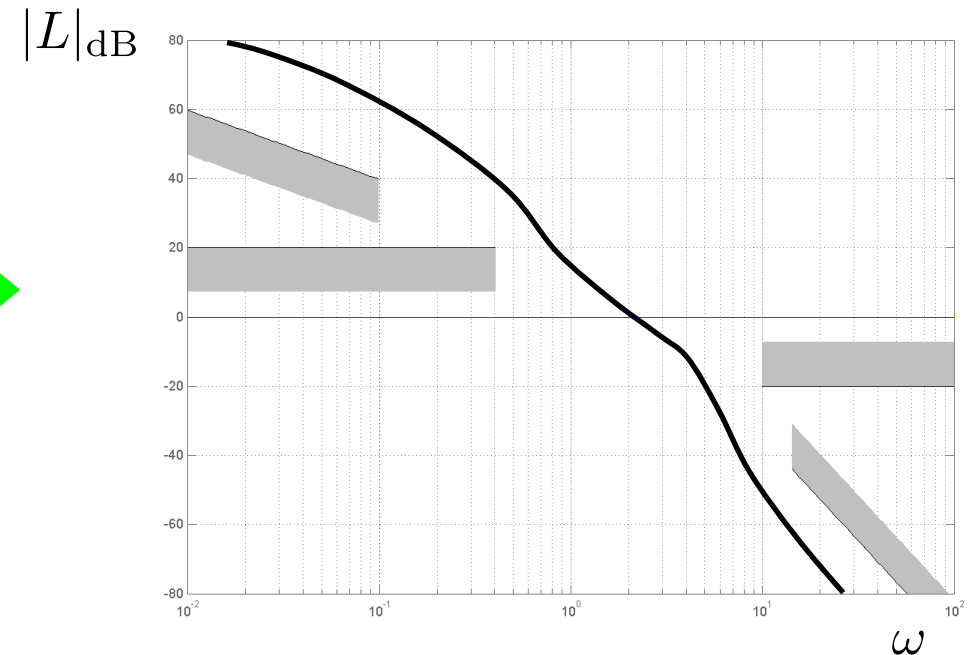
$$|e(\infty)| \leq \bar{e}, \text{ with } w, d \text{ specified}$$

- Dynamic Specifications

- $\omega_{\min} \leq \omega_c \leq \omega_{\max}$
- $\varphi_m \geq \bar{\varphi}_m$
- $K_m \geq \bar{K}_m$



## Constraints on open-loop Bode diagrams of $|L(j\omega)|_{\text{dB}}$



# Design Specifications on Bode Diagrams: Example

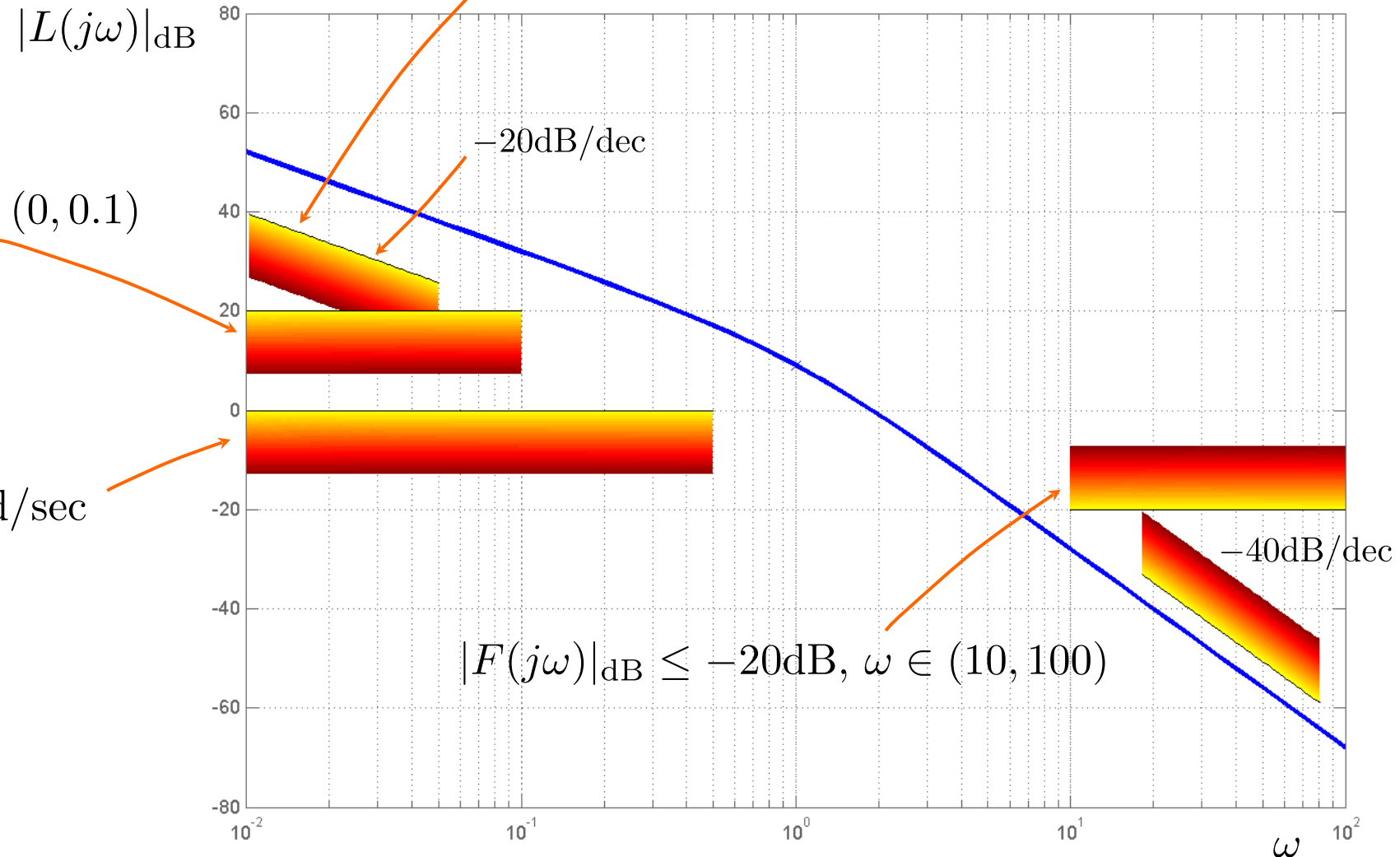
$$L(s) = \frac{4}{s(s+1)}$$

$$|e(\infty)| = 0, \text{ with } w(t) = A \cdot 1(t), \forall A$$

$$|S(j\omega)|_{\text{dB}} \leq -20\text{dB}, \omega \in (0, 0.1)$$

$$\Delta\% \leq 3 \quad \longrightarrow \quad \xi \geq 0.75$$

$$t_s \leq 10 \text{ sec} \quad \longrightarrow \quad \omega_c \geq 0.5 \text{ rad/sec}$$



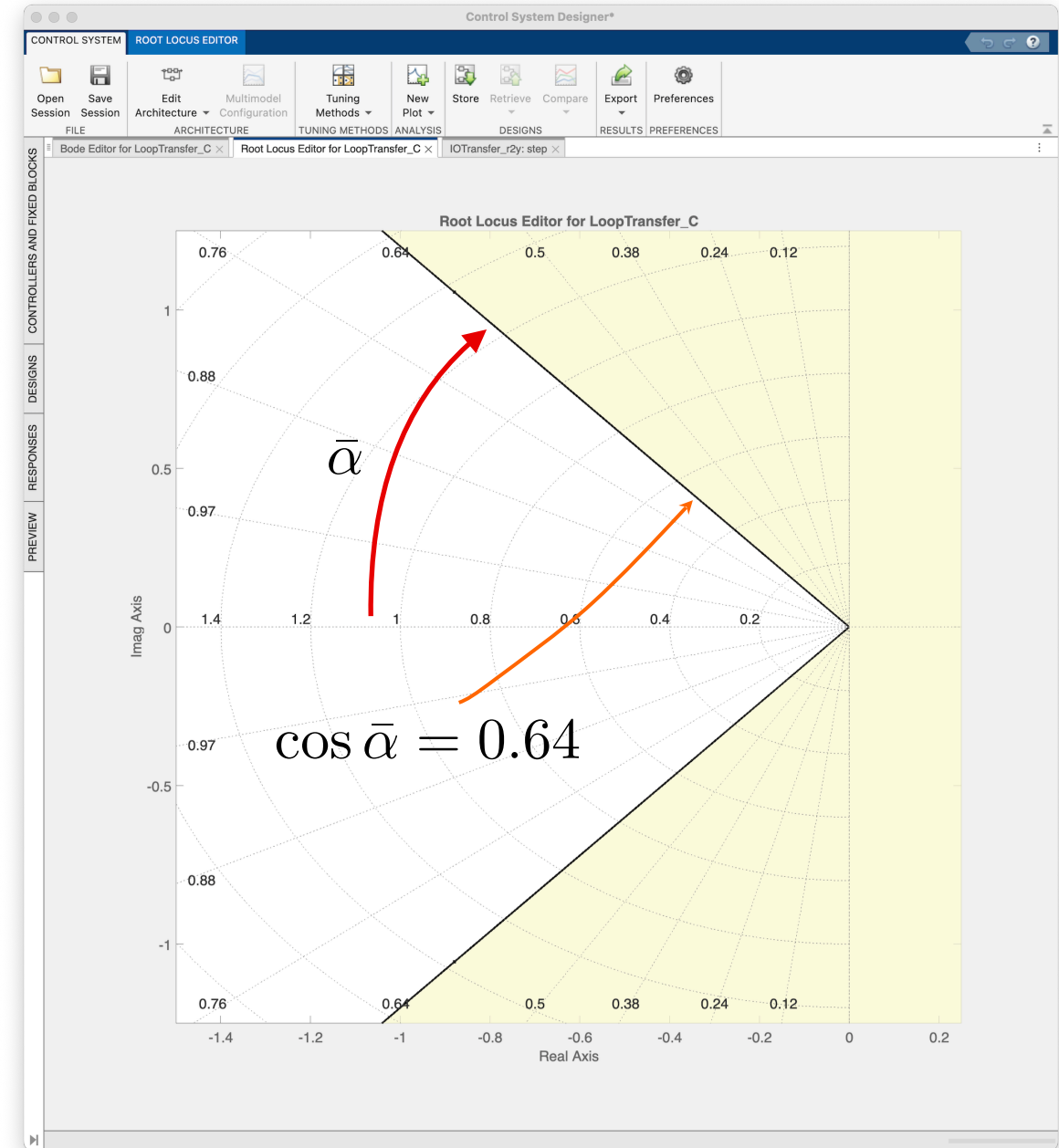
Closed-loop requirement:

- the **damping ratio**  $\xi$  has to satisfy the constraint

$$\xi \geq \bar{\xi}$$

- The region of the complex plane where the constraint  $\xi \geq \bar{\xi}$  is satisfied can be drawn as a **graphical constraint** on the RL:

$$s \in \mathbb{C} : \xi = \cos \alpha \geq \bar{\xi} = \cos \bar{\alpha}$$



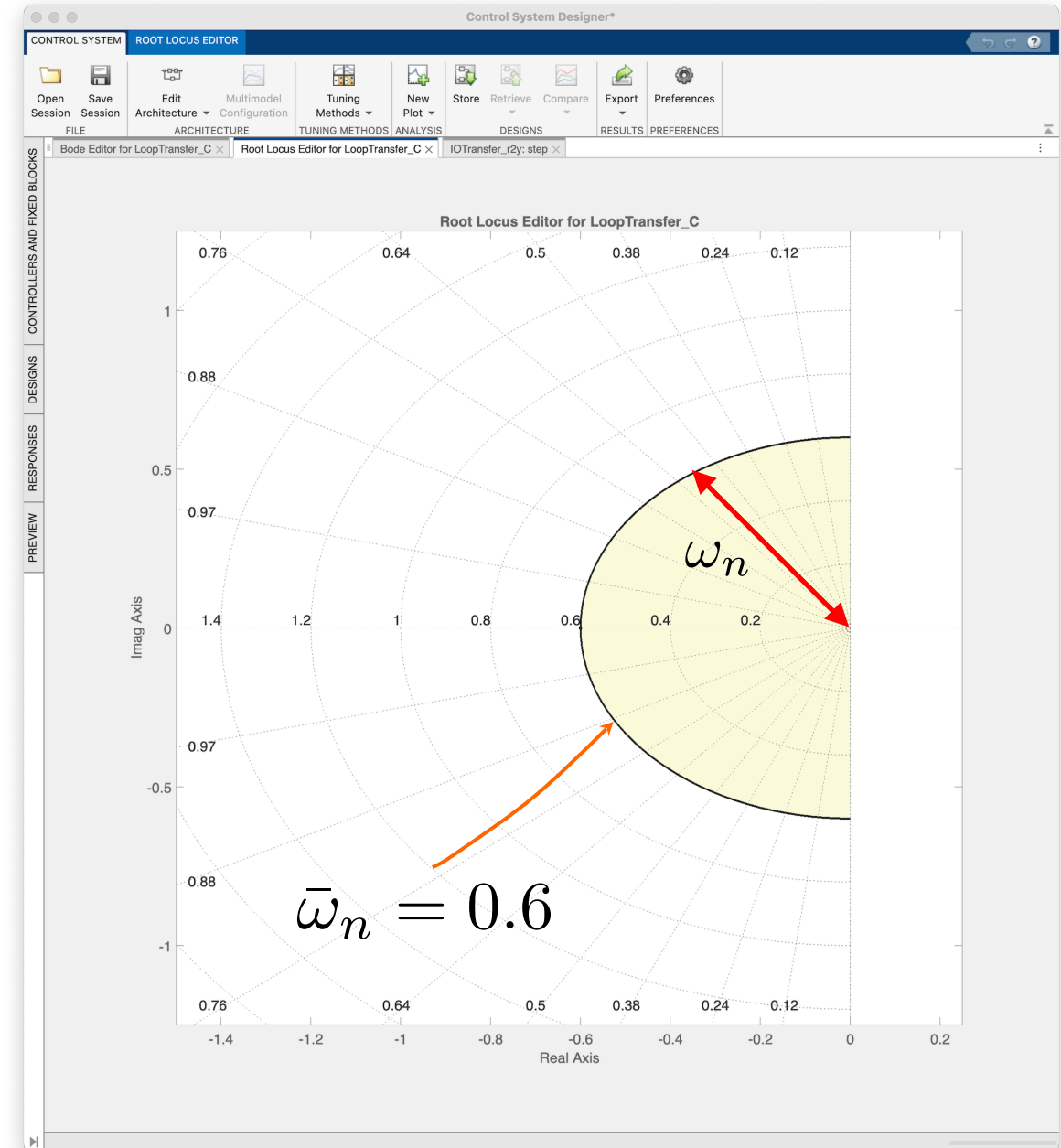
Closed-loop requirement:

- the **natural angular frequency**  $\omega_n$  has to satisfy the constraint

$$\omega_n \geq \bar{\omega}_n$$

- The region of the complex plane where the constraint  $\omega_n \geq \bar{\omega}_n$  is satisfied can be drawn as a **graphical constraint** on the RL:

$$s \in \mathbb{C} : \omega_n \geq \bar{\omega}_n$$



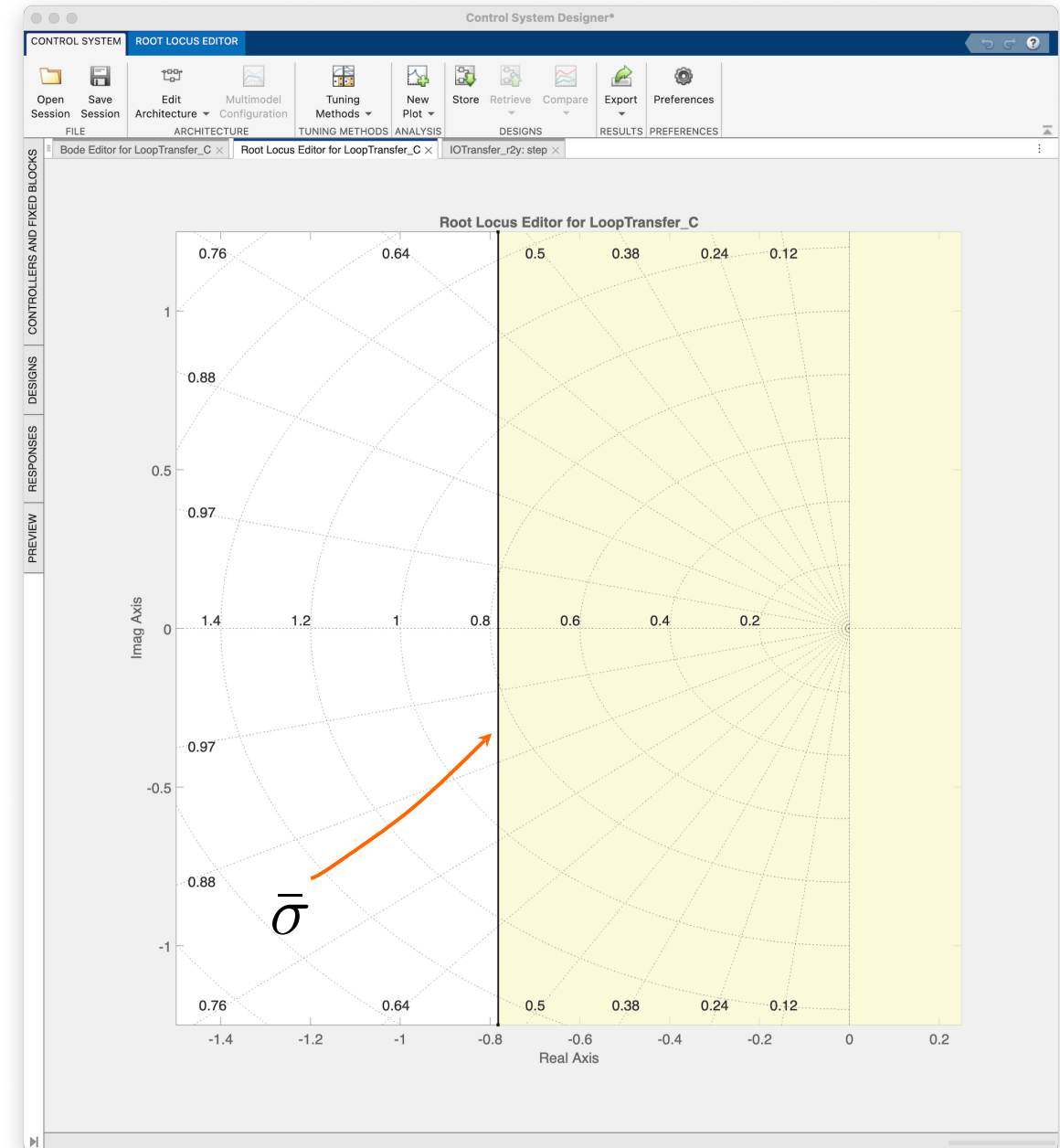
Closed-loop requirement:

- the (negative) **real part of the poles**  $\sigma$  has to satisfy the constraint

$$\sigma \leq \bar{\sigma}$$

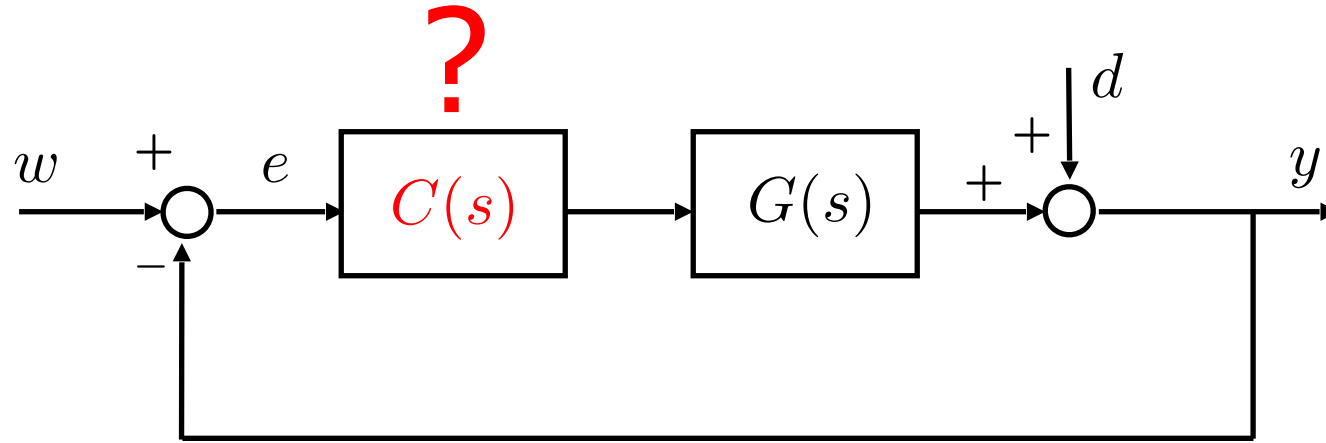
- The region of the complex plane where the constraint  $\sigma \leq \bar{\sigma}$  is satisfied can be drawn as a **graphical constraint** on the RL:

$$s \in \mathbb{C} : \sigma = \operatorname{Re}(s) \leq \bar{\sigma}$$



- Analyse the **closed-loop requirements** and translate them to **design specifications on the Bode diagrams**
- Select an **initial attempt** for the controller  $C(s)$  using a very simple structure. For example, a typical initial choice is an algebraic controller  $C_0(s) = \mu_C$
- Check whether the chosen controller satisfies **all specifications**. If not, consider another attempt using a more complex structure of the controller and proceed with the design trying to meet all specifications
- **Iterate** the procedure till when all specifications are met

# Loop-Shaping Iterative Controller Design: Example 1



$$G(s) = \frac{10}{(1 + 10s)(1 + 5s)(1 + s)}$$

- **Design specifications:**

- $|e(\infty)| \leq 0.1$  with  $\begin{cases} w(t) = A \cdot 1(t), & |A| \leq 1 \\ d(t) = B \cdot 1(t), & |B| \leq 5 \end{cases}$
- $\omega_c \geq 0.2$
- $\varphi_m \geq 60^\circ$

- **Controller Structure:**

$$C(s) = C_1(s) \cdot C_2(s) \quad \text{where} \quad C_1(s) = \frac{\mu_C}{sg}; \quad C_2(s) = \frac{\prod (1 + sT_i)}{\prod (1 + s\tau_i)}$$

used to meet **static**  
specifications



used to meet **dynamic**  
specifications





$$L(s) = C(s)G(s) = C_1(s)C_2(s) \frac{10}{(1 + 10s)(1 + 5s)(1 + s)}$$

## Static Design:

Since  $C_2(0) = 1$  the **static design is not influenced by**

gain:  $10\mu_C > 0$   
type:  $g$

Due to the **linearity** of the closed-loop system:

$$e(\infty) = e_w(\infty) + e_d(\infty)$$

where:

$$e_w(t) = e(t) \big|_{d(t)=0} ; \quad e_d(t) = e(t) \big|_{w(t)=0}$$

Hence:

$$e(\infty) = e_w(\infty) + e_d(\infty)$$

  $|e(\infty)| \leq |e_w(\infty)| + |e_d(\infty)|$

and:

$$|e_w(\infty)| = \begin{cases} \frac{|A|}{1 + 10\mu_C} \leq \frac{1}{1 + 10\mu_C}, & \text{if } g = 0 \\ 0, & \text{if } g > 0 \end{cases}$$

$$|e_d(\infty)| = \begin{cases} \frac{|B|}{1 + 10\mu_C} \leq \frac{5}{1 + 10\mu_C}, & \text{if } g = 0 \\ 0, & \text{if } g > 0 \end{cases}$$

- **Option 1:** static design with  $g = 0$  (no poles in the origin introduced by  $C_1(s)$ )

$$|e(\infty)| \leq \frac{1}{1 + 10\mu_C} + \frac{5}{1 + 10\mu_C} = \frac{6}{1 + 10\mu_C}$$

Imposing  $|e(\infty)| \leq 0.1$  :

$$\downarrow \frac{6}{1 + 10\mu_C} \leq 0.1 \quad \longrightarrow \mu_C \geq 5.9$$

Possible choice:  $C_1(s) = 8$

- **Option 2:** static design with  $g = 1$  (one pole in the origin introduced by  $C_1(s)$ )

$$\downarrow e(\infty) = 0, \quad \forall \mu_C \quad \longrightarrow \quad C_1(s) = \frac{\mu_C}{s}$$

## Dynamic Design, Option (A):

Pick the static controller:  $C_1(s) = 8$

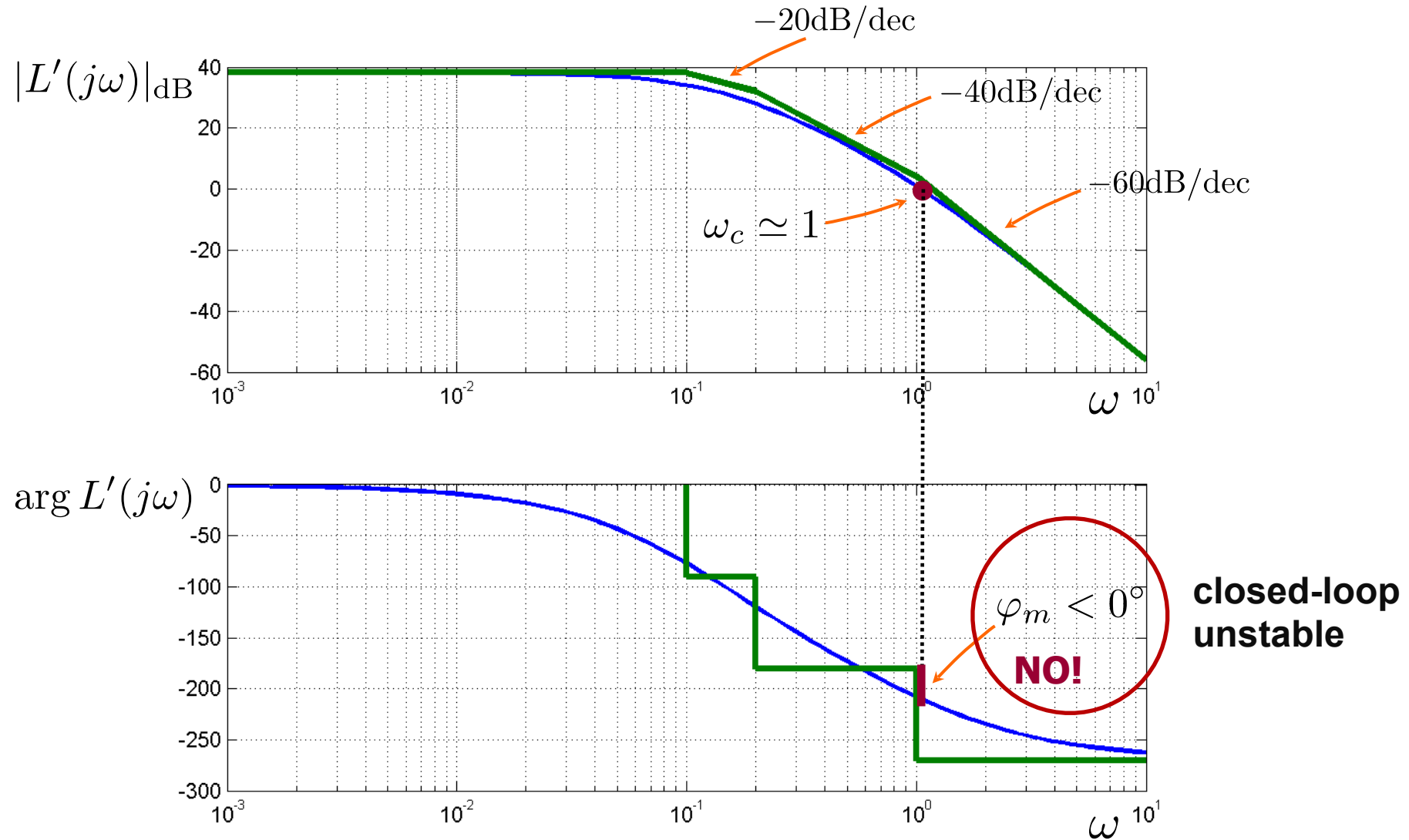
↳  $L(s) = 8 G(s) C_2(s) = L'(s) C_2(s)$

$$\text{with } L'(s) = \frac{80}{(1 + 10s)(1 + 5s)(1 + s)}$$

We proceed by a **sequence of attempts** with increasing complexity of the structure of controller  $C_2(s)$  by following the **iterative procedure** outlined in [Slide 8](#):

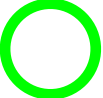
# Loop-Shaping Iterative Controller Design: Example 1 (contd.)

**Attempt 1:**  $C_1(s) = 8$ ,  $C_2(s) = 1$

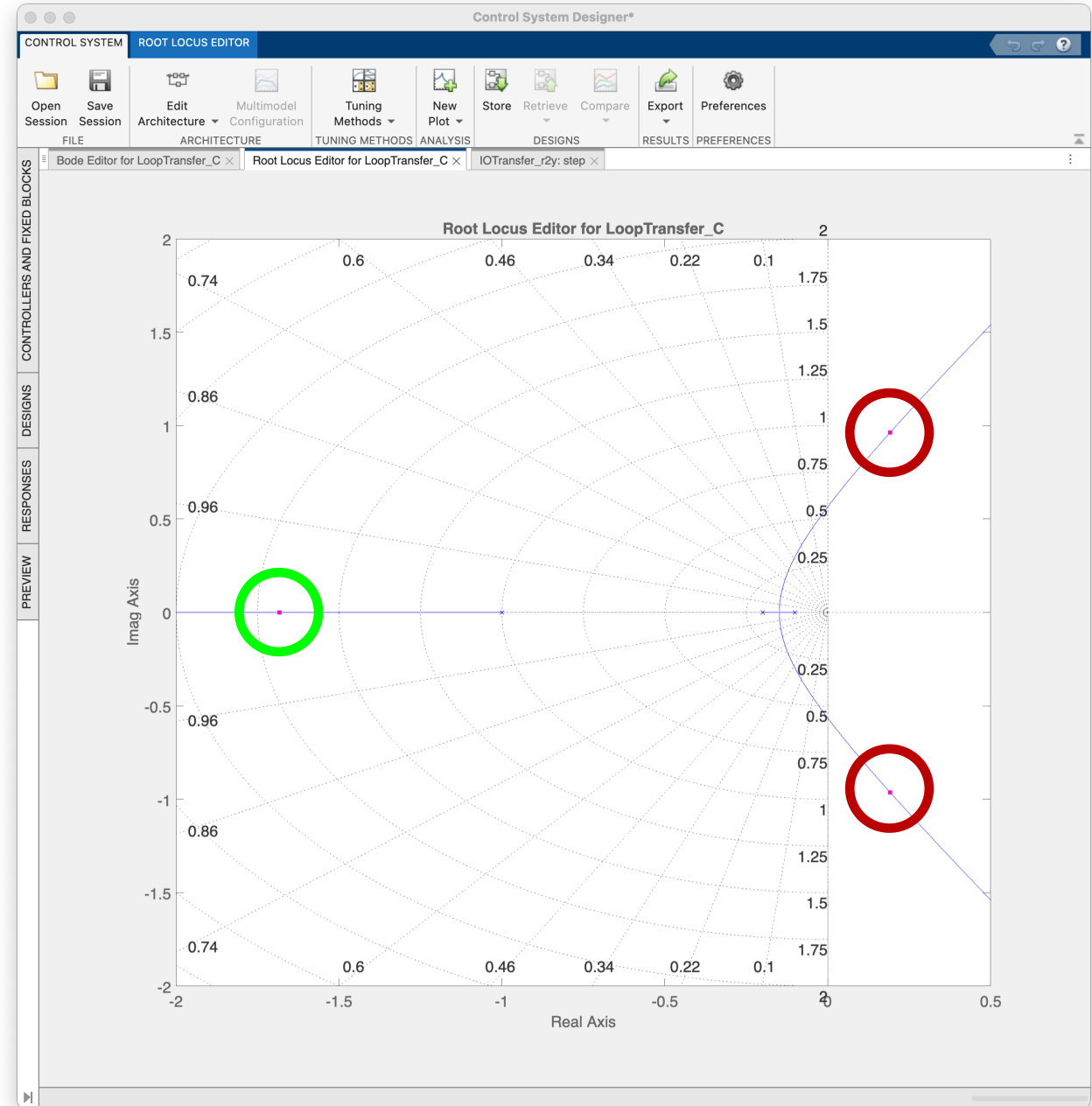


**Attempt 1:**  $C_1(s) = 8$ ,  $C_2(s) = 1$

 unstable **closed-loop** pole

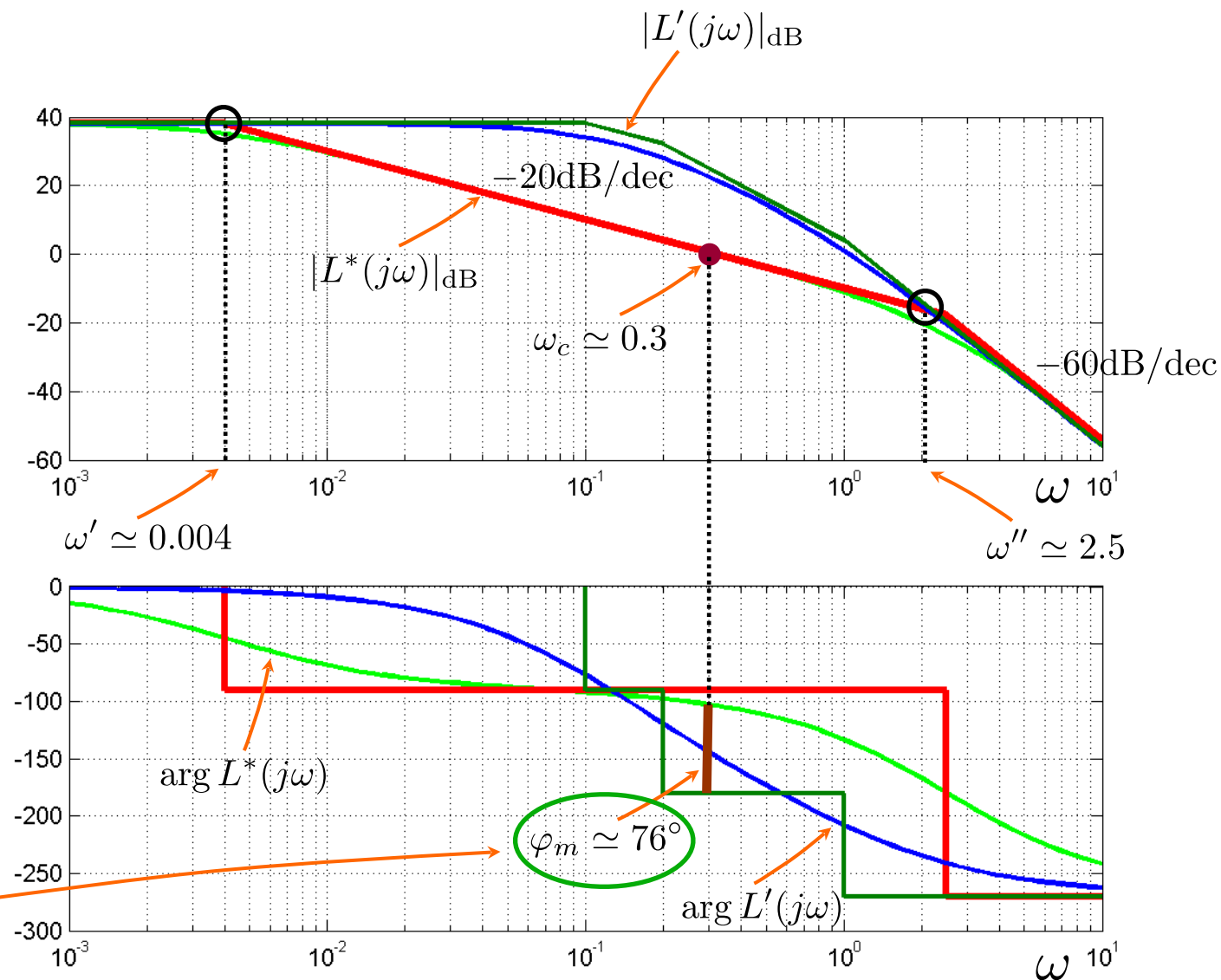
 asymptotically stable **closed-loop** pole

- The choice of controller  $C_1(s) = 8$  to satisfy the static requirement is **not feasible** because the closed-loop system would be **unstable**
- Decreasing the gain in  $C_1(s)$  would "*move*" the close-loop poles into the open left-plane but this way the static requirement would **not** be satisfied



## Attempt 2:

closed-loop  
asympt. stable



## Procedure followed in Attempt 2 of Dynamic Design Option (A) :

- Set a satisfactory critical angular frequency:  $\omega_c = 0.3 \geq 0.2$
- Draw a straight line on the magnitude Bode plot with slope  $-20\text{dB/dec}$  and passing through the point  $(\omega_c, 0\text{dB})$  ; find the frequencies  $\omega'$ ,  $\omega''$  where the straight line intersects the diagram of  $|L'(j\omega)|_{\text{dB}}$
- Construct the asymptotic Bode diagram of  $|L^*(j\omega)|_{\text{dB}}$  which coincides with the drawn straight line for  $\omega \in [\omega', \omega'']$  and with the diagram of  $|L'(j\omega)|_{\text{dB}}$  for  $\omega < \omega'$  and  $\omega > \omega''$
- The resulting asymptotic diagram of  $|L^*(j\omega)|_{\text{dB}}$  gives:

$$L^*(s) = \frac{80}{(1 + s/0.004)(1 + s/2.5)^2} \quad \longrightarrow \quad C_2(s) = \frac{L^*(s)}{L'(s)} = \frac{(1 + 10s)(1 + 5s)(1 + s)}{(1 + 250s)(1 + 0.4s)^2}$$



## Analysis of the controller obtained with Design Option (A):

guarantees the **required static precision**

zeros introduced to “cancel” the poles  $-0.1, -0.2, -1$  thus **increasing**  $\arg L(j\omega)$

$$C(s) = C_1(s)C_2(s) = 8 \frac{(1 + 10s)(1 + 5s)(1 + s)}{(1 + 250s)(1 + 0.4s)^2}$$

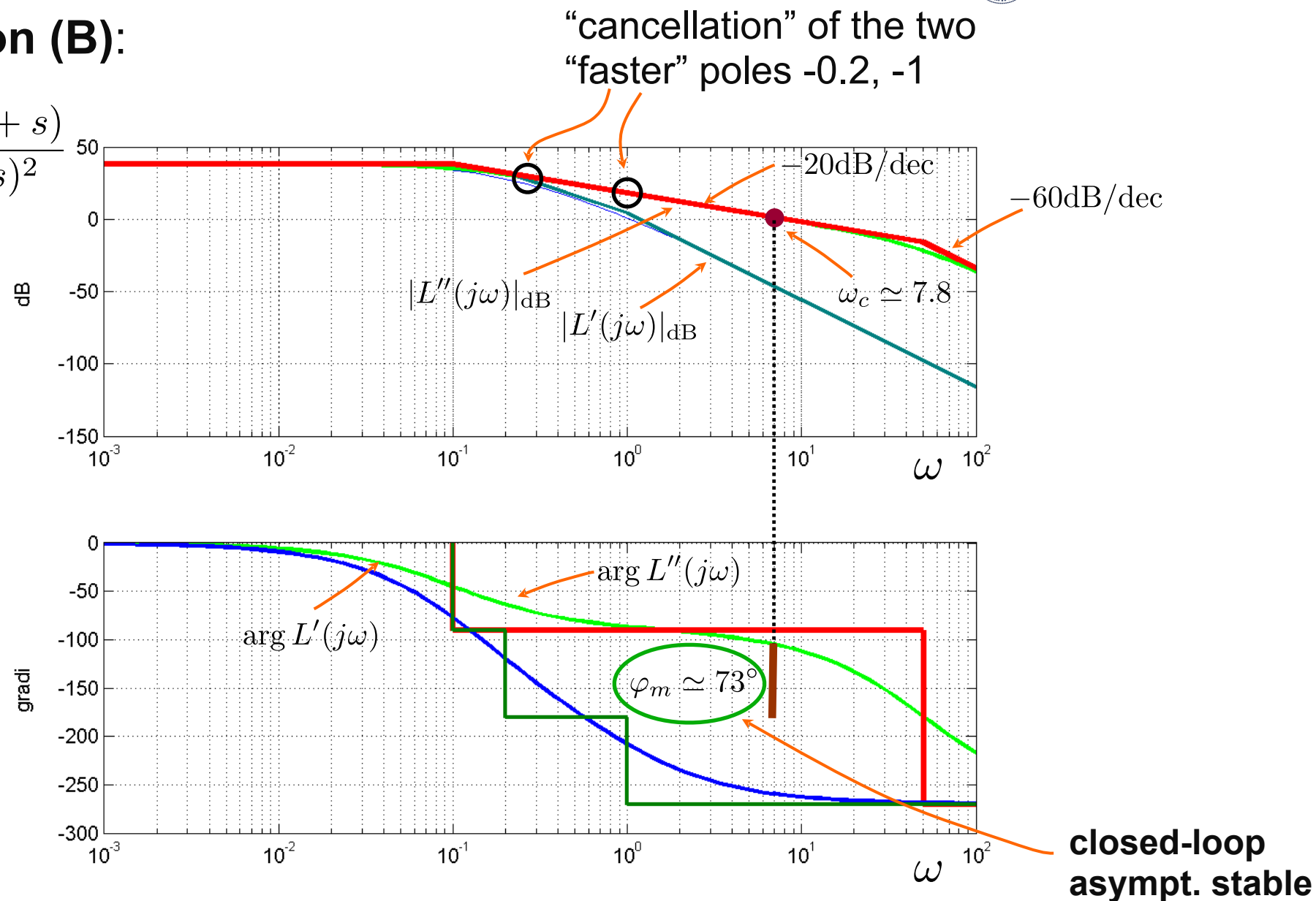
“moves” to low frequency the slope of  $-20\text{dB/dec}$  in order to cross the  $0\text{dB}$  axis at the desired critical frequency with a slope  $-20\text{dB/dec}$  to guarantee the desired phase margin (the system is minimum phase and hence **the Bode criterion can be used**)

### high-frequency poles:

- physical realisability
- attenuation of high-frequency output disturbances

## Dynamic Design, Option (B):

$$C(s) = 8 \frac{(1 + 10s)(1 + 5s)(1 + s)}{(1 + 250s)(1 + 0.4s)^2}$$



## Dynamic Design, Option (B):

$$C(s) = 8 \frac{(1 + 10s)(1 + 5s)(1 + s)}{(1 + 250s)(1 + 0.4s)^2}$$

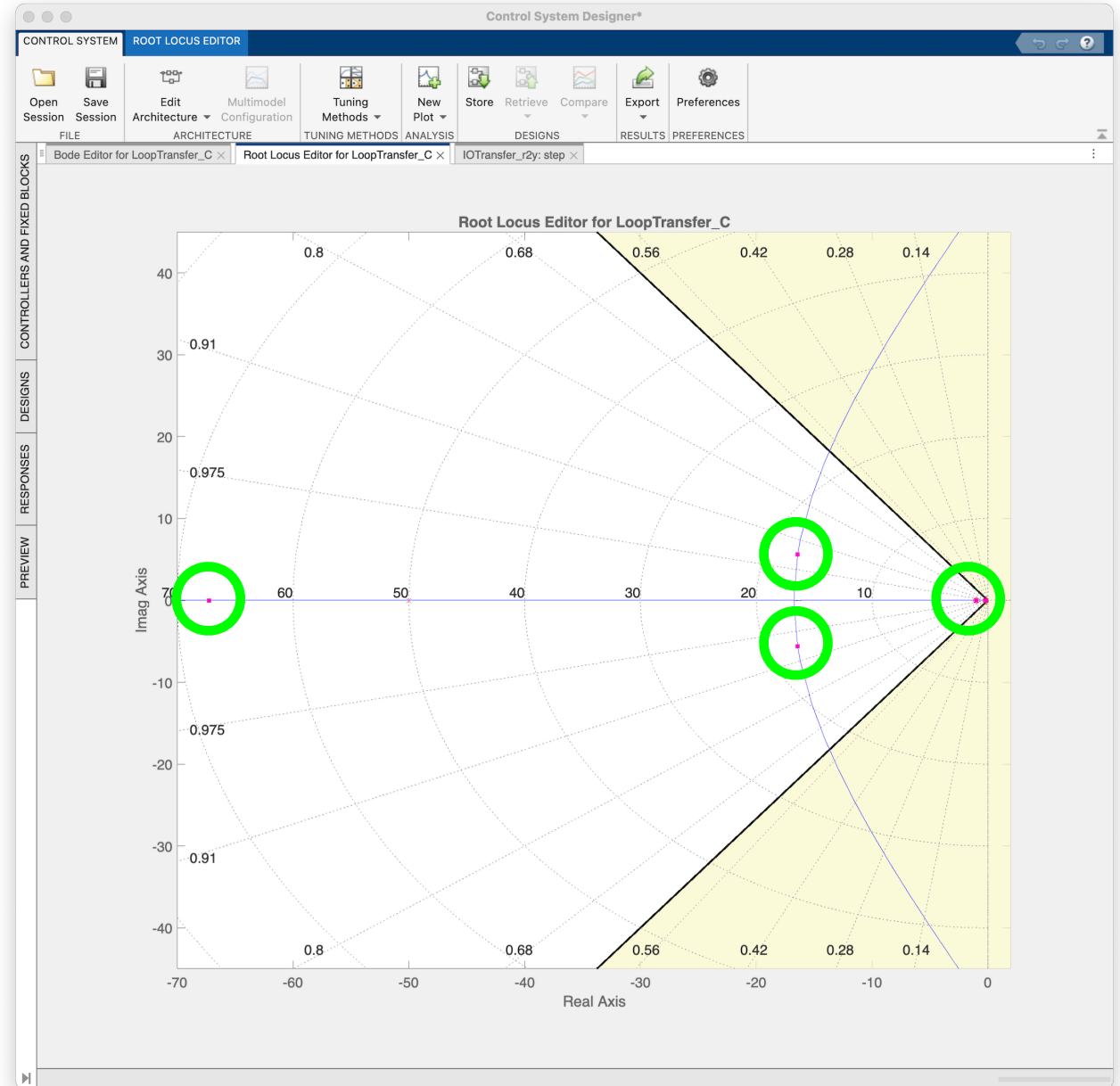
### ○ closed-loop pole

- All **closed-loop** poles are asymptotically stable.
- From the Requirement  $\omega_c \geq 0.2$  we obtain the **RL graphical constraint**

$$\omega_n \simeq \omega_c \geq 0.2$$

- From the Requirement  $\varphi_m \geq 60^\circ$  we obtain the **RL graphical constraint**

$$\xi \simeq \frac{\varphi_m}{100} \geq \frac{60^\circ}{100} = 0.6$$



## Procedure followed in Option (B) :

- The static design is kept unchanged with  $C_1(s) = 8$
- The “slower” pole in -0.1 is kept whereas the “faster” poles in -0.2, -1 are cancelled by respective zeros introduced by the controller to ensure a slope  $-20\text{dB/dec}$  when crossing the point  $(\omega_c, 0\text{dB})$
- Two sufficiently fast (at frequency sufficiently larger than  $\omega_c$ ) poles are introduced by the controller
- The resulting asymptotic diagram of  $|L''(j\omega)|_{\text{dB}}$  gives:

$$L''(s) = \frac{80}{(1 + 10s)(1 + 0.02s)^2} \quad \longrightarrow \quad C_2(s) = \frac{L''(s)}{L'(s)} = \frac{(1 + 5s)(1 + s)}{(1 + 0.02s)^2}$$

## Static and Dynamic Design, Option (C):

Pick the static controller (thus we also modify the static design):  $C_1(s) = \frac{\mu_C}{s}$

↳ 
$$L(s) = \frac{G(s)}{s} \mu_C C_2(s) = L_1(s) \mu_C C_2(s)$$

$$\text{with } L_1(s) = \frac{10}{s(1 + 10s)(1 + 5s)(1 + s)}$$

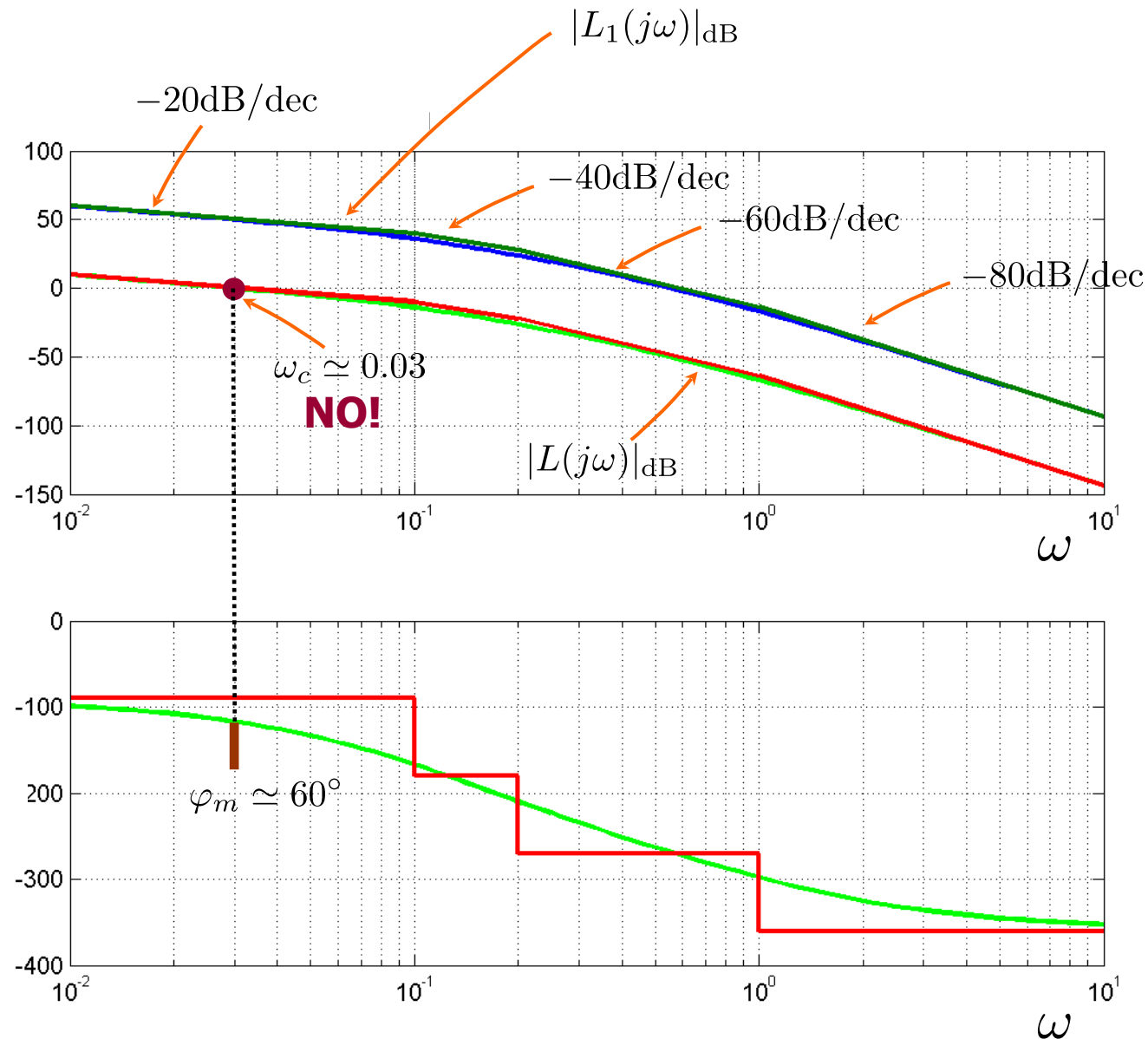
Again, we proceed by a **sequence of attempts** with increasing complexity of the structure of controller  $C_2(s)$  by following the **iterative procedure** outlined in [Slide 8](#):

## Attempt 1:

$$C_2(s) = 1$$

$$\rightarrow L(s) = \mu_C L_1(s)$$

$$\rightarrow \mu_C = 10^{-5/2}$$



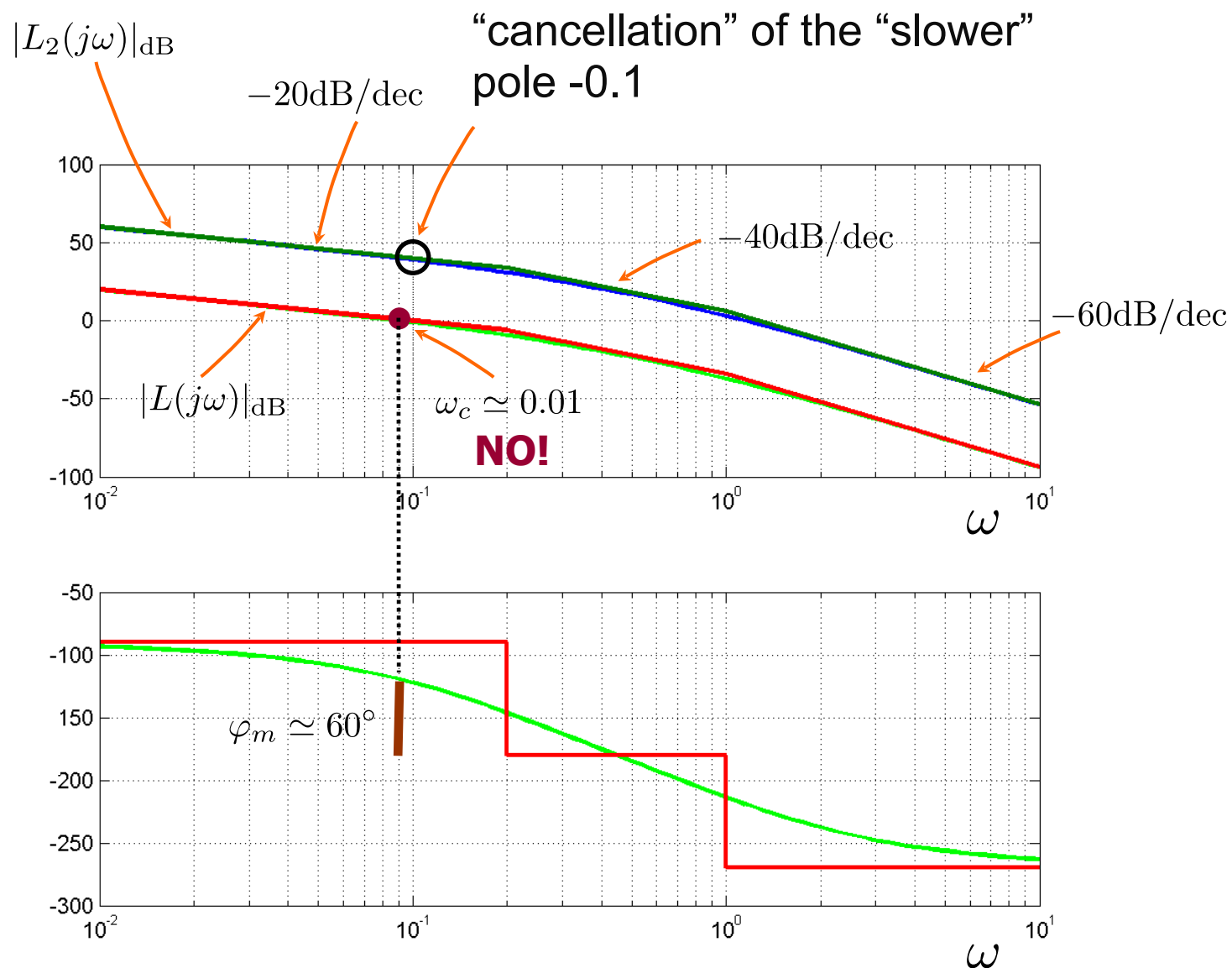
## Attempt 2:

$$C_2(s) = (1 + 10s)$$

$$L_2(s) = \frac{10}{s(1 + 5s)(1 + s)}$$

$$L(s) = \mu_C L_2(s)$$

$$\mu_C \simeq 0.01$$



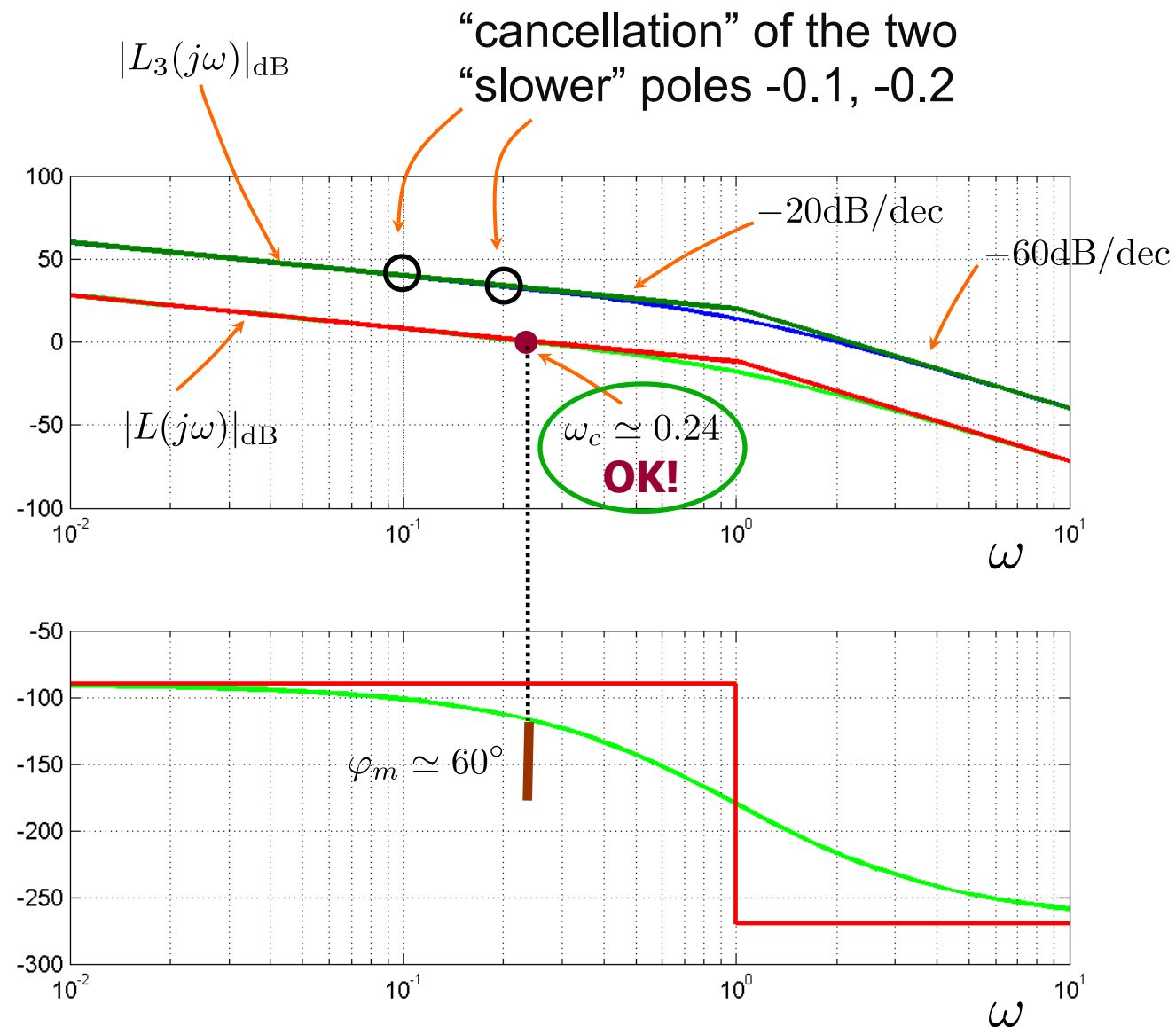
## Attempt 3:

$$C_2(s) = \frac{(1 + 10s)(1 + 5s)}{1 + s}$$

$$\rightarrow L_3(s) = \frac{10}{s(1 + s)^2}$$

$$\rightarrow L(s) = \mu_C L_3(s)$$

$$\rightarrow \mu_C \simeq 0.0254$$





## Dynamic Design, Option (C):

$$C(s) = 0.0254 \frac{(1 + 10s)(1 + 5s)}{s(1 + s)}$$

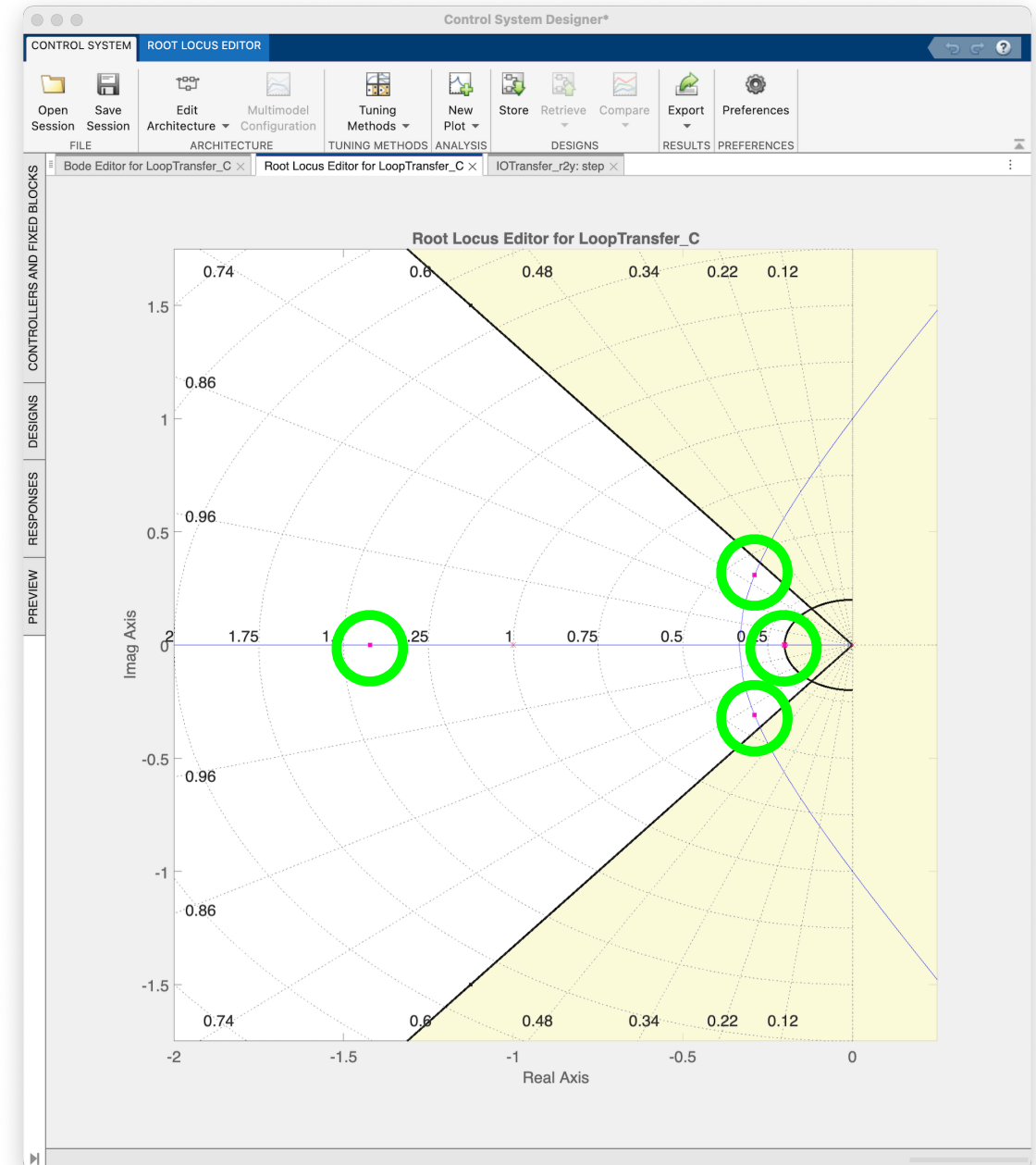
 **closed-loop pole**

- All **closed-loop** poles are asymptotically stable.
- From the Requirement  $\omega_c \geq 0.2$  we obtain the **RL graphical constraint**

$$\omega_n \simeq \omega_c \geq 0.2$$

- From the Requirement  $\varphi_m \geq 60^\circ$  we obtain the **RL graphical constraint**

$$\xi \simeq \frac{\varphi_m}{100} \geq \frac{60^\circ}{100} = 0.6$$



## Example 1: Comparison Between the Designed Controllers

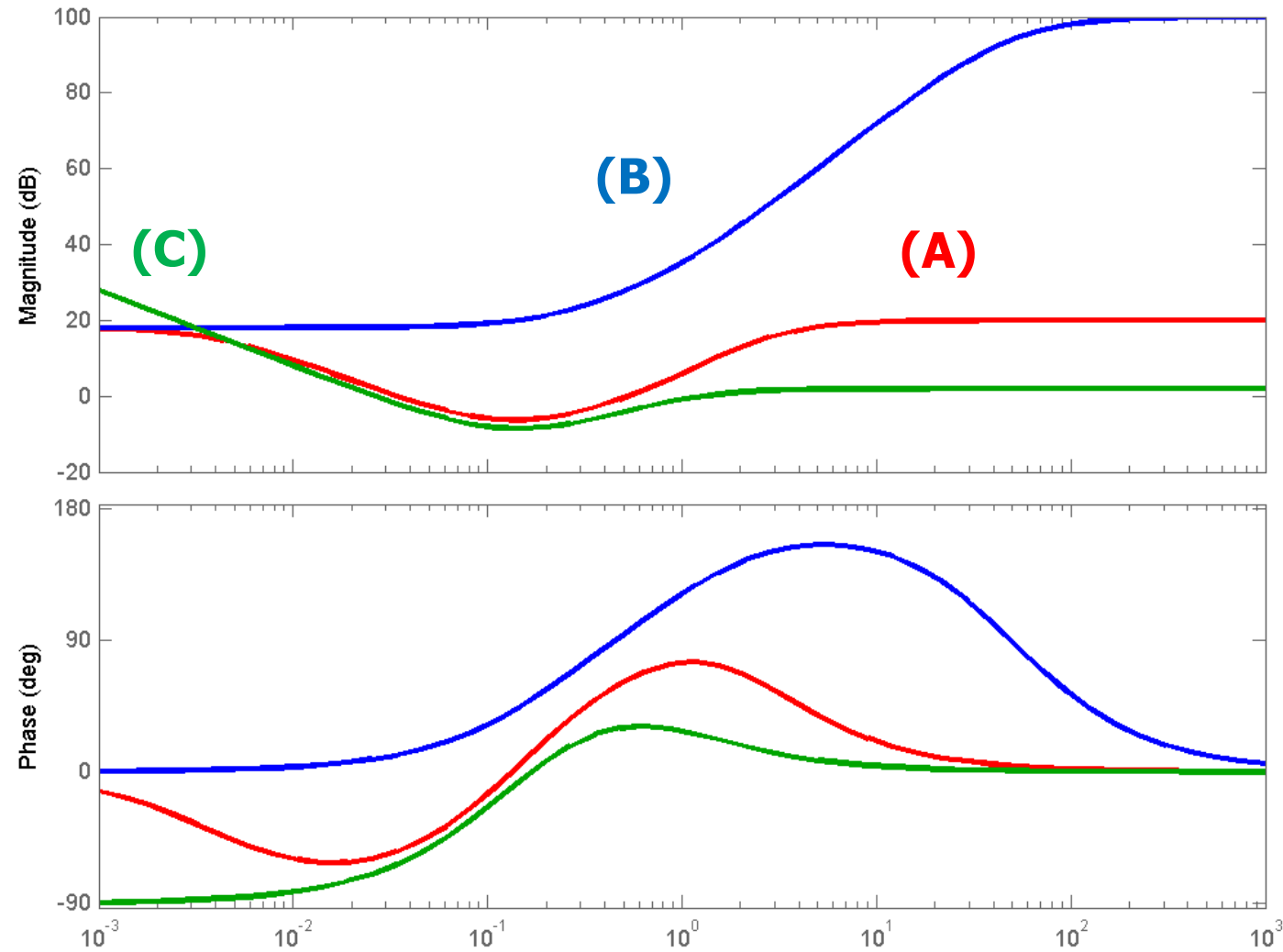
$$G(s) = \frac{10}{(1 + 10s)(1 + 5s)(1 + s)}$$

- **Design specifications:**

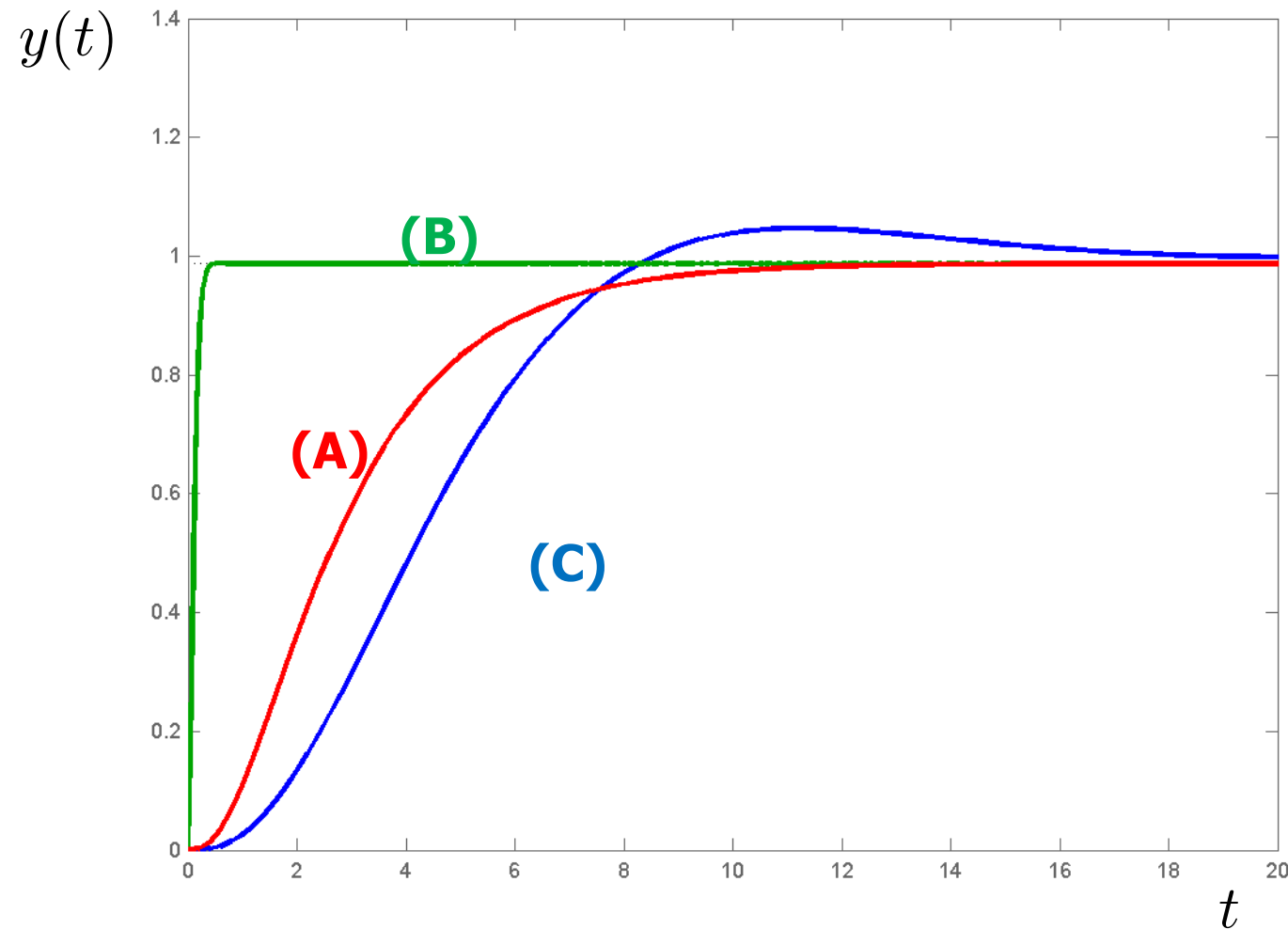
- $|e(\infty)| \leq 0.1$  with  $\begin{cases} w(t) = A \cdot 1(t), & |A| \leq 1 \\ d(t) = B \cdot 1(t), & |B| \leq 5 \end{cases}$
- $\omega_c \geq 0.2$
- $\varphi_m \geq 60^\circ$

|                                                                                |                                                                                                                                                                                                          |                     |
|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>(A):</b> $C(s) = 8 \frac{(1 + 10s)(1 + 5s)(1 + s)}{(1 + 250s)(1 + 0.4s)^2}$ | $\left\{ \begin{array}{l} \omega_c \simeq 0.3 \\ \varphi_m \simeq 76^\circ \\ \omega_c \simeq 7.8 \\ \varphi_m \simeq 73^\circ \\ \omega_c \simeq 0.24 \\ \varphi_m \simeq 63^\circ \end{array} \right.$ | and $e(\infty) = 0$ |
| <b>(B):</b> $C(s) = 8 \frac{(1 + 5s)(1 + s)}{(1 + 0.02s)^2}$                   |                                                                                                                                                                                                          |                     |
| <b>(C):</b> $C(s) = 0.0254 \frac{(1 + 10s)(1 + 5s)}{s(1 + s)}$                 |                                                                                                                                                                                                          |                     |

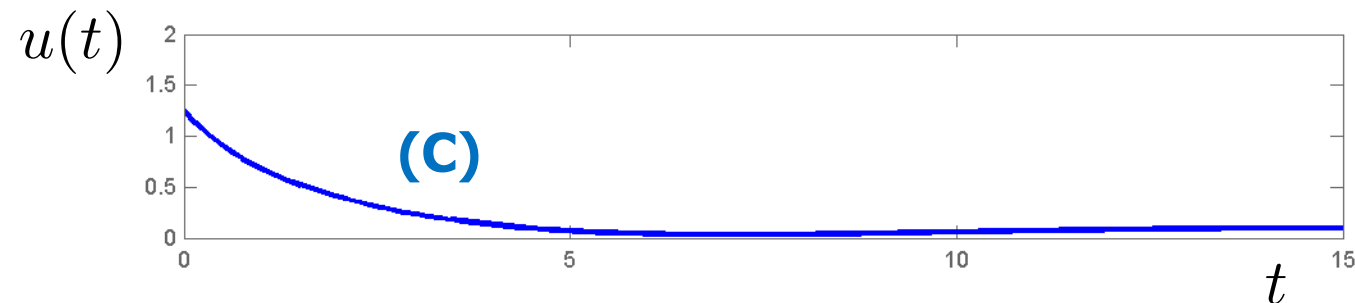
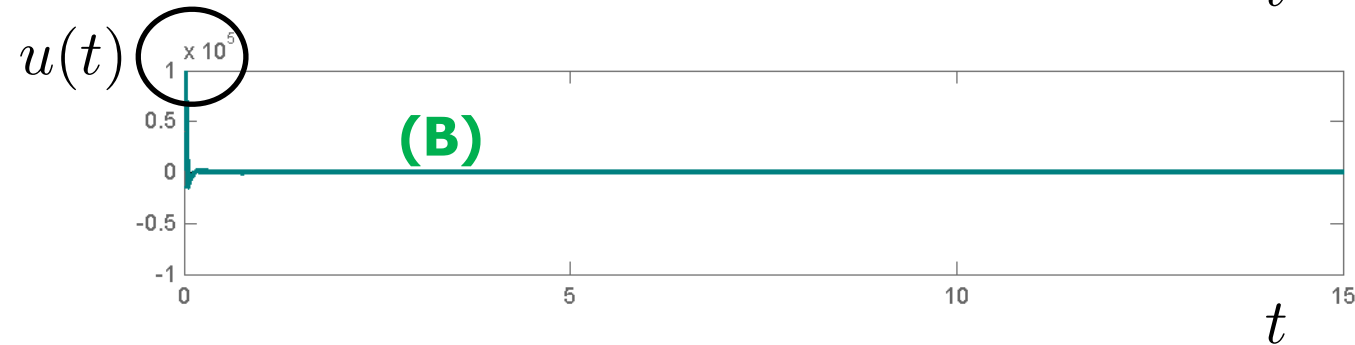
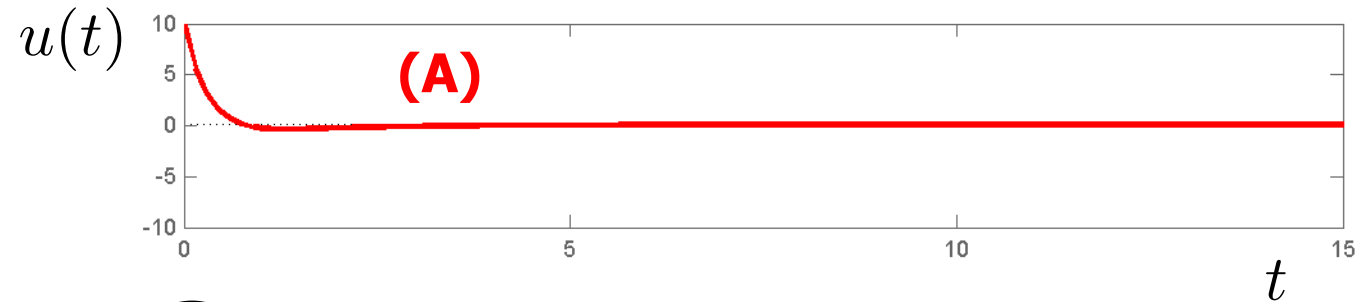
## Bode diagrams of the controllers



## Closed-loop step response



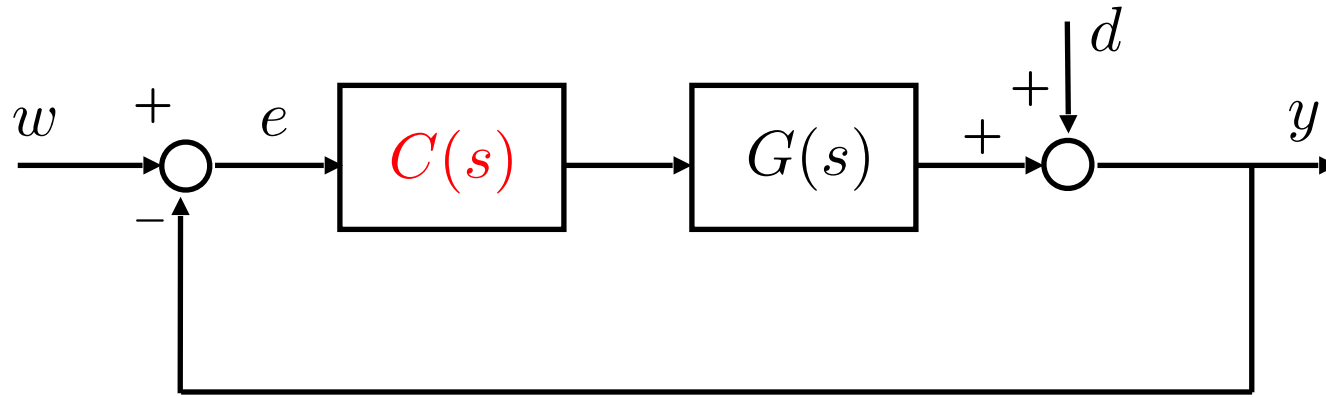
## Closed-loop step response



In scenarios where the system to be controlled has **non-minimum phase zeros** or is **unstable**:

- the **Bode criterion** cannot be used
- the “**cancellation**” of zeros and poles with non-negative real part is **not allowed** not to generate unstable hidden dynamics
- the presence of zeros and/or poles with non-negative real part introduces **inherent limitations on the achievable performance**

## Example 2 - System with a Non-Minimum Phase Zero



$$G(s) = \frac{10(1 - 2s)}{s(1 + 10s)(1 + 0.1s)}$$

**Design specifications:**  $\varphi_m \geq 40^\circ$

The term  $\frac{(1 - 2s)}{s}$  :

- cannot be “canceled”
- it gives a **negative contribution** to the phase:  $-90^\circ - \text{arctg}(2\omega)$

↳ for  $\omega = 0.5$  the negative contribution to the phase is  $-135^\circ$

↳ the critical frequency  $\omega_c$  cannot exceed "too much"  $\omega_c = 0.5$

↳ **strong limitation to the achievable speed of response**

## Example 2 - System with a Non-Minimum Phase Zero (contd.)

Note that:

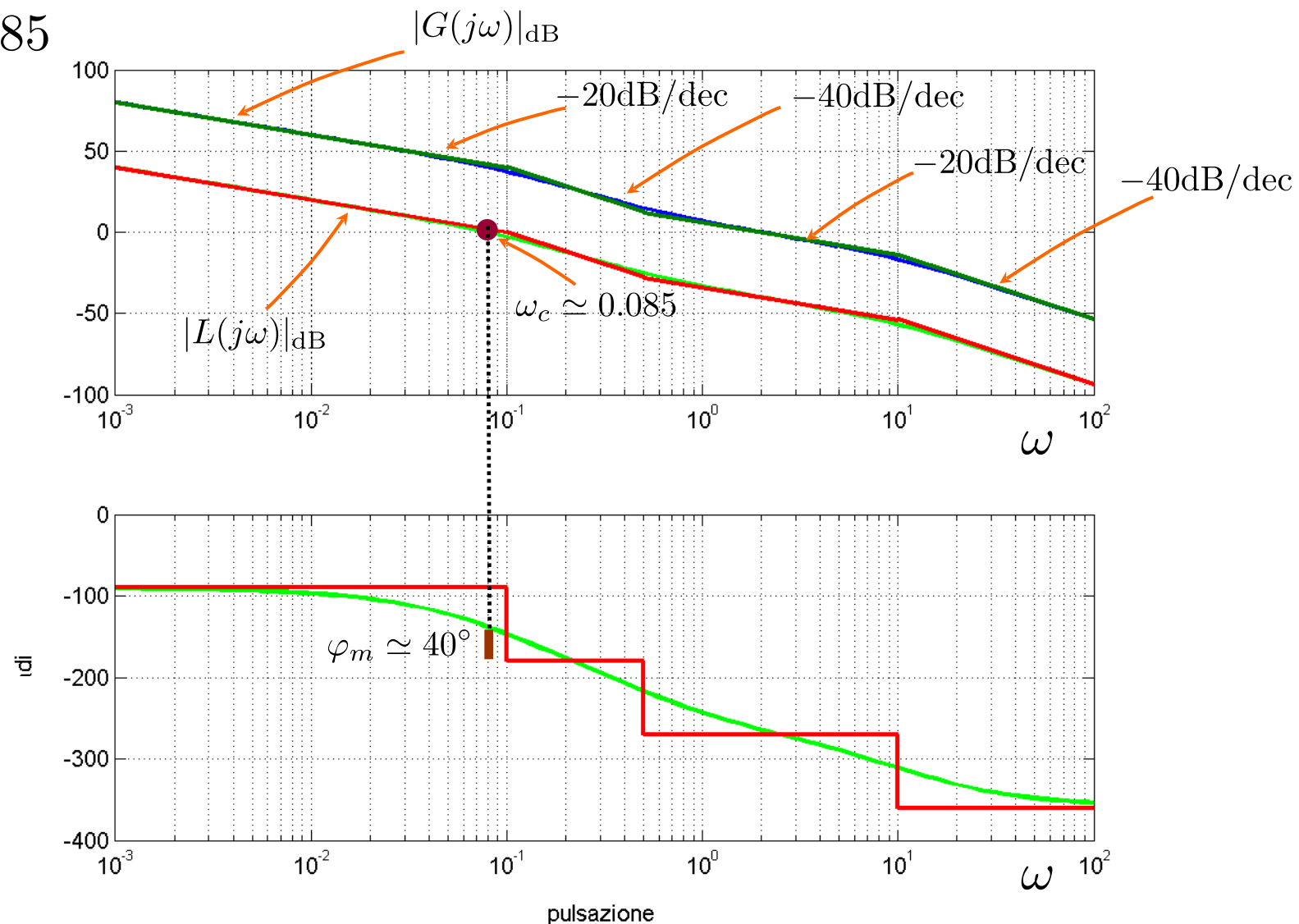
$$\arg G(j\omega) = -140^\circ \text{ for } \omega \simeq 0.085$$

**Attempt 1:**

$$C(s) = \mu_C ; L(s) = \mu_C G(s)$$

→  $\omega_{c\max} = 0.085$

→  $\mu_C \simeq 0.01$





## Attempt 2:

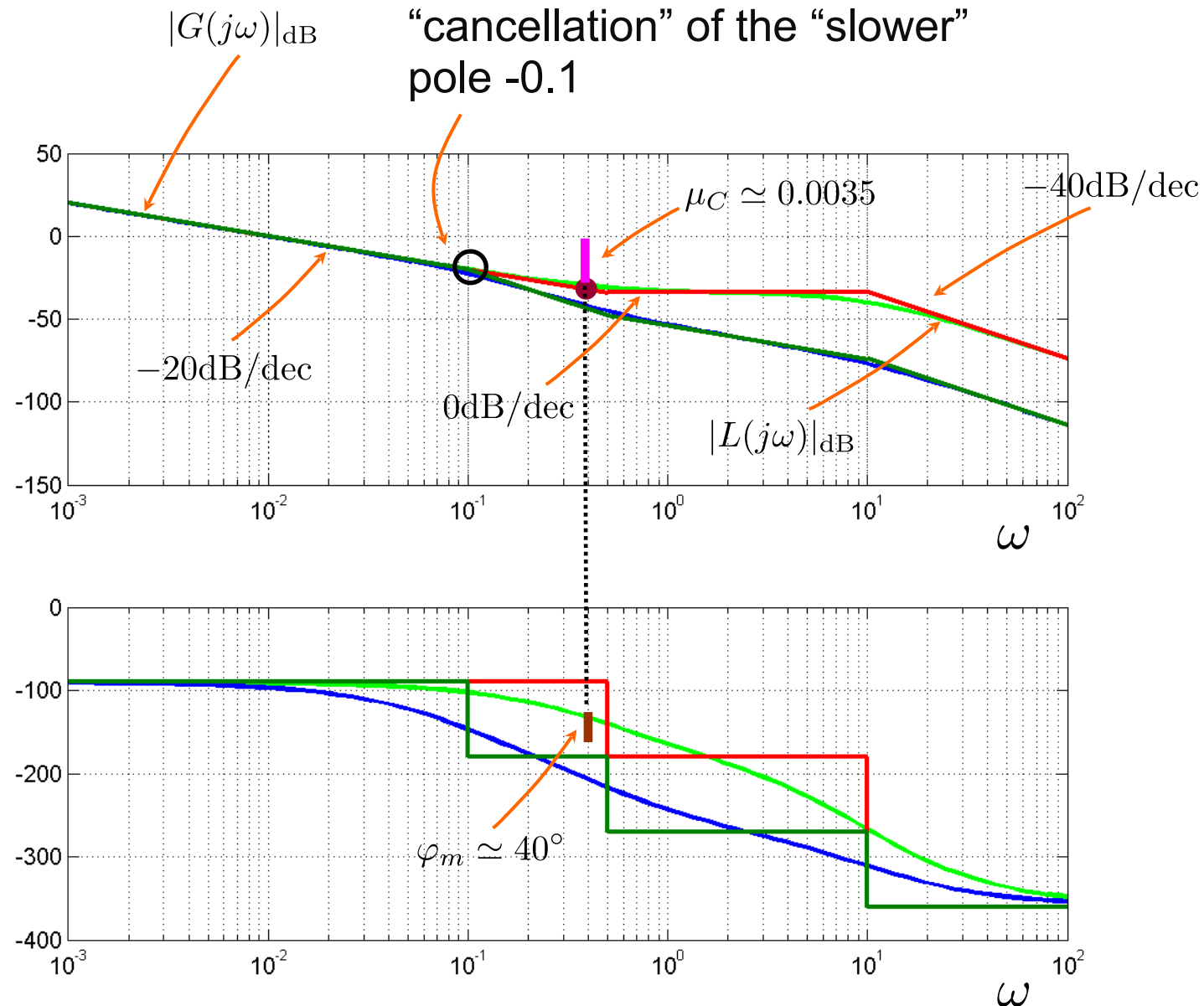
$$C(s) = \mu_C \frac{1 + 10s}{1 + 0.1s}$$

↳ 
$$L(s) = \mu_C \frac{10(1 - 2s)}{s(1 + 0.1s)^2}$$

Note that:

$$\arg L(j\omega) = -140^\circ \text{ for } \omega \simeq 0.5$$

↳ 
$$\mu_C \simeq 0.0035$$



## Dynamic Design, Attempt 2:

$$C(s) = 0.0035 \frac{1 + 10s}{1 + 0.1s}$$

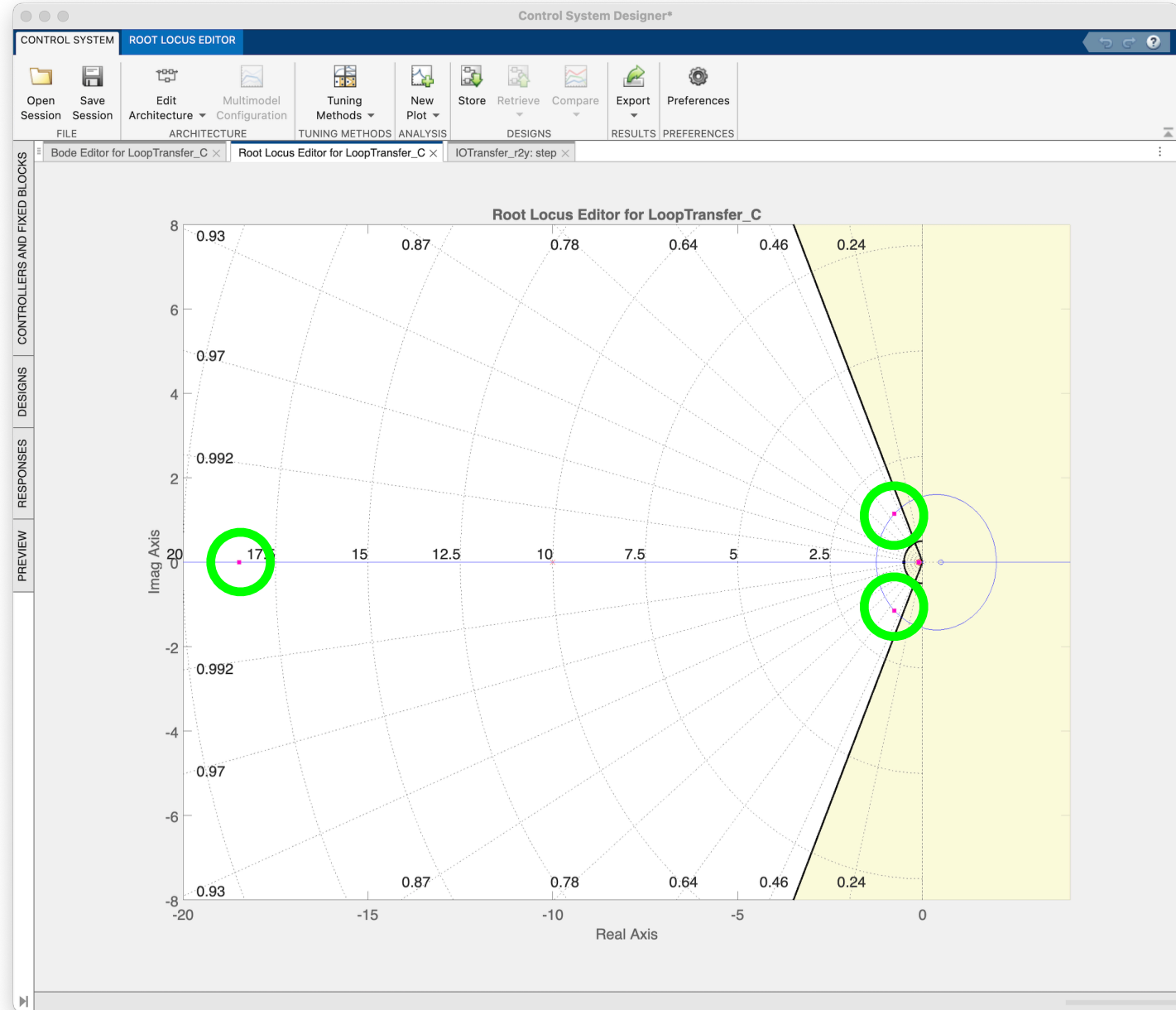
 **closed-loop pole**

- All **closed-loop** poles are asymptotically stable.
- From the Requirement  $\varphi_m \geq 40^\circ$  we obtain the **RL graphical constraint**

$$\xi \simeq \frac{\varphi_m}{100} \geq \frac{40^\circ}{100} = 0.4$$

- Even if it is not a requirement, to get a controller that "pushes" the performance to the limits, we impose  $\omega_c \geq 0.5$ , thus we obtain the **RL graphical constraint**

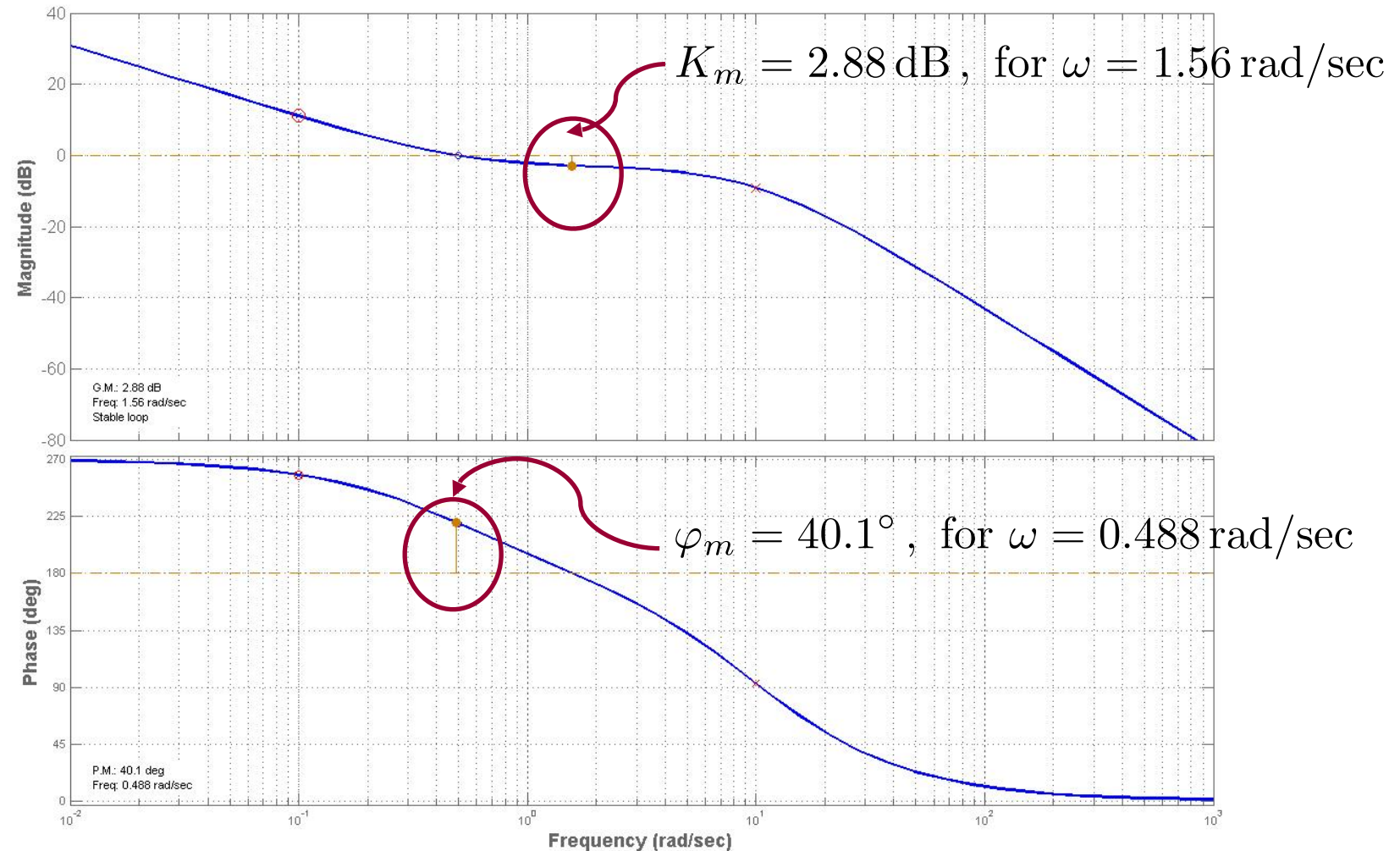
$$\omega_n \simeq \omega_c \geq 0.5$$



## Example 2 - System with a Non-Minimum Phase Zero (contd.)

Actual performance with controller obtained by Attempt 2:

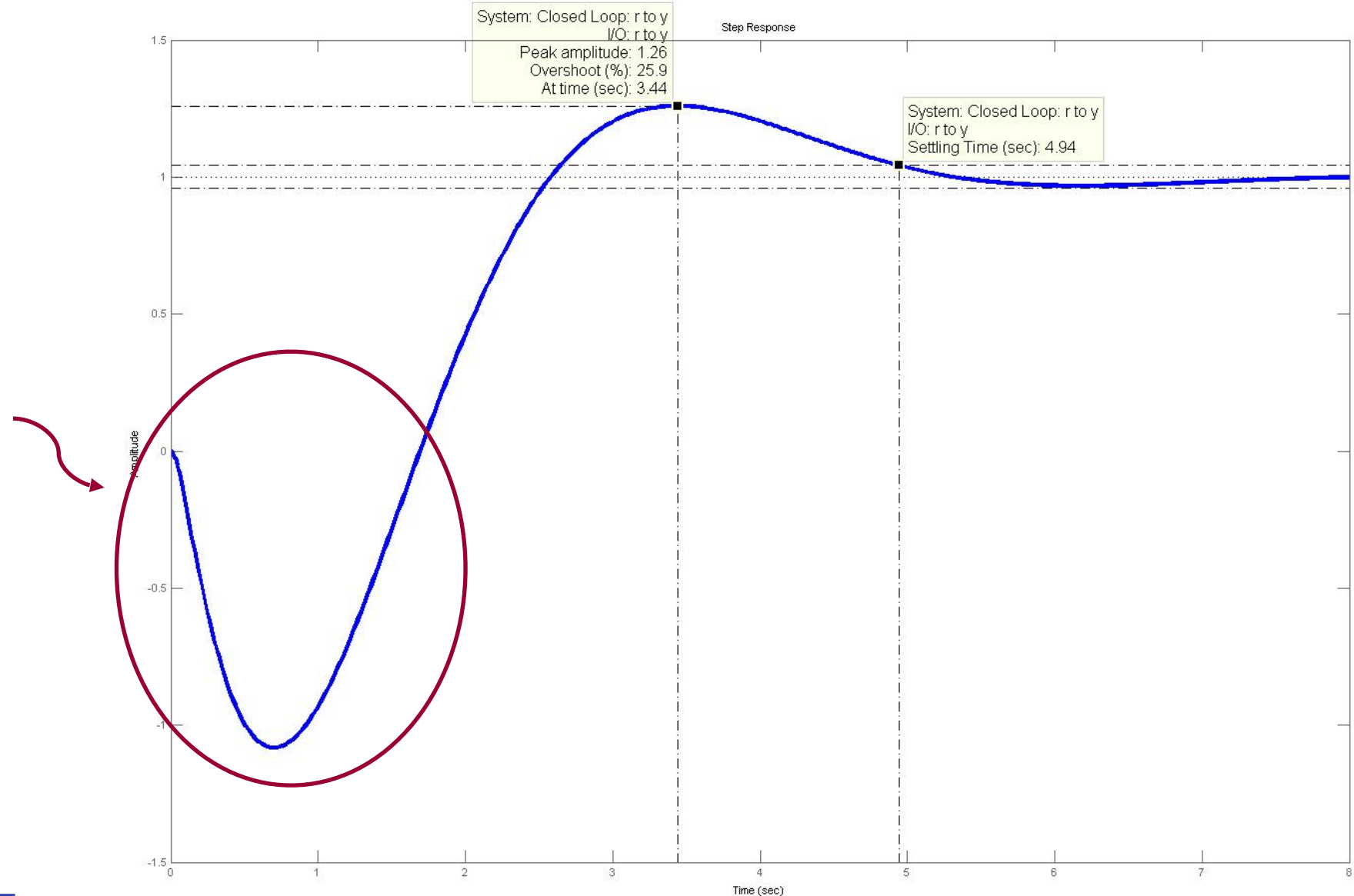
Note that the **gain margin**  $K_m$  is rather small



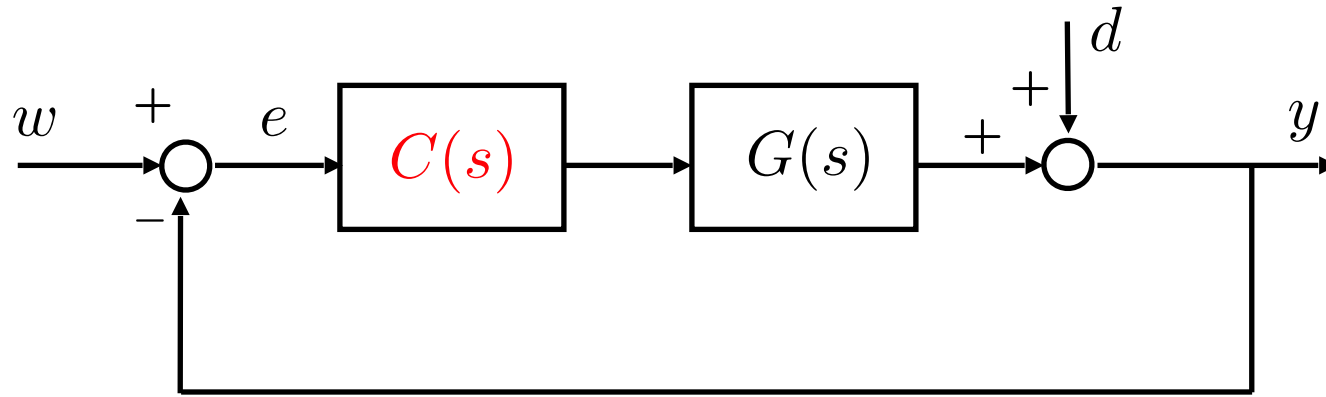
## Example 2 - System with a Non-Minimum Phase Zero (contd.)

### Actual performance with controller obtained by Attempt 2:

Also, note the  
rather  
significant  
**under-shoot**  
(typical for non-  
minimum phase  
systems)



## Example 3 - Unstable System

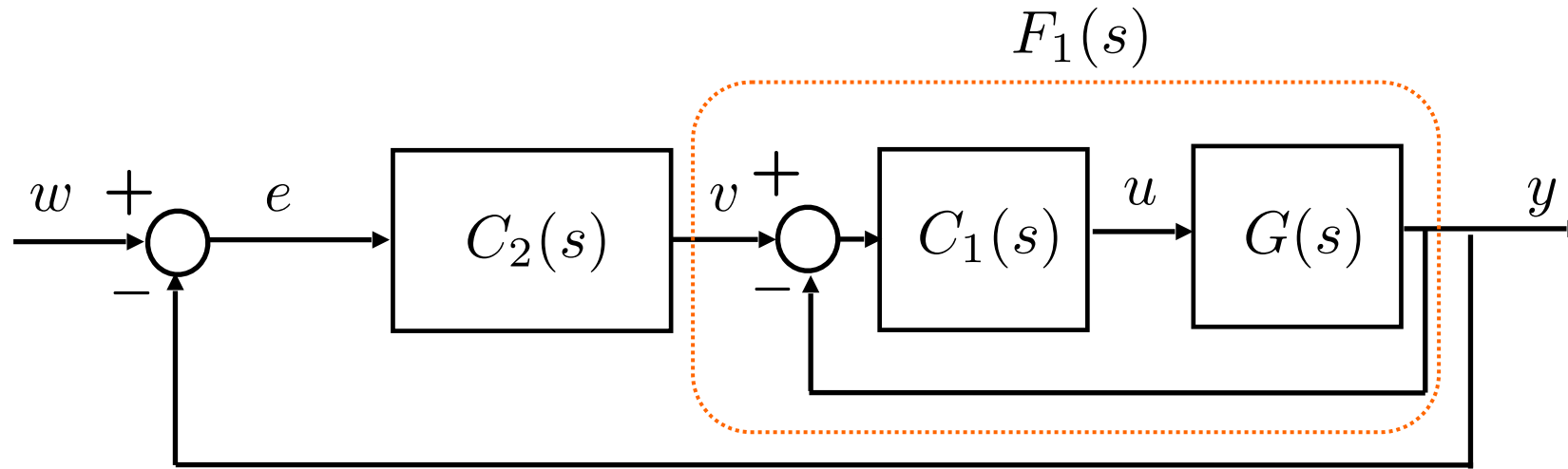


$$G(s) = \frac{1}{s - 1}$$

### Design specifications:

- $|e(\infty)| = 0$  with  $\begin{cases} w(t) = A \cdot 1(t), \forall A \\ d(t) = 0, \forall t \end{cases}$
- $\omega_c \geq 0.5$
- $\varphi_m \geq 45^\circ$

The simplest and frequently used approach is to consider a **dual-loop controller architecture**



- Controller  $C_1(s)$  is designed to **stabilise** the system described by the inner closed-loop transfer function  $F_1(s)$
- Controller  $C_2(s)$  is designed to **meet the specifications** for the whole feedback control system

The design of the inner controller  $C_1(s)$  can be carried out as:

$$C_1(s) = \mu_1 \quad \longrightarrow \quad L_1(s) = C_1(s) \cdot G(s) = \frac{\mu_1}{s - 1}$$

$$\quad \searrow \quad F_1(s) = \frac{L_1(s)}{1 + L_1(s)} = \frac{\mu_1}{s - 1 + \mu_1}$$

By choosing (for example):  $\mu_1 = 11$

$$\quad \searrow \quad F_1(s) = \frac{1.1}{1 + 0.1s} \quad \text{asymptotically stable}$$

## Example 3 - Unstable System (contd.)

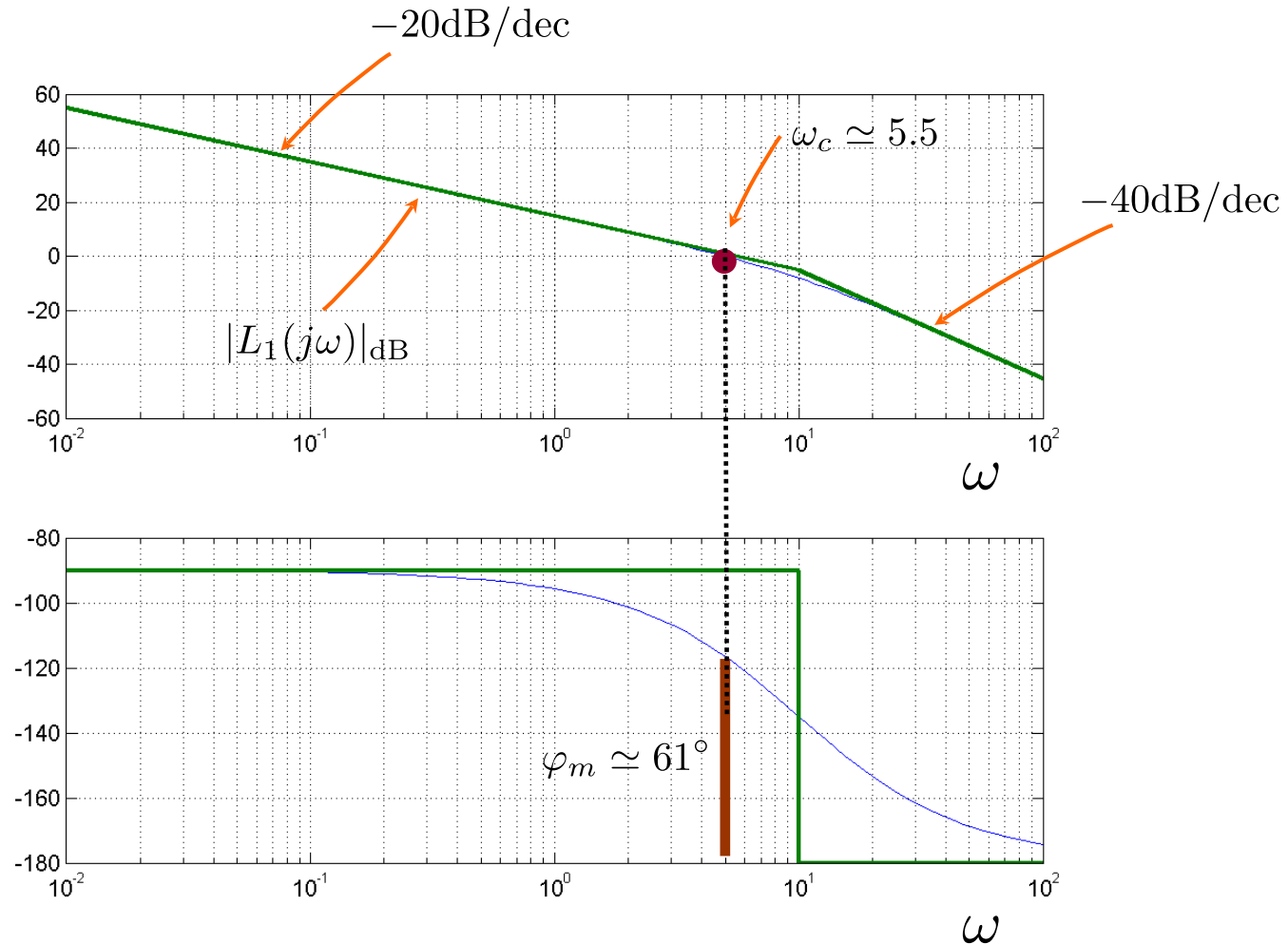
- Now, the design of the outer controller  $C_2(s)$  is very simple:
- To meet the specification on the static error, an **integrator** on the direct path has to be introduced.
- Hence, by choosing:

$$C_2(s) = \frac{5}{s}$$

↳  $L_1(s) = C_2(s) \cdot F_1(s)$

$$= \frac{5.5}{s(1 + 0.1s)}$$

↳ All specifications are met

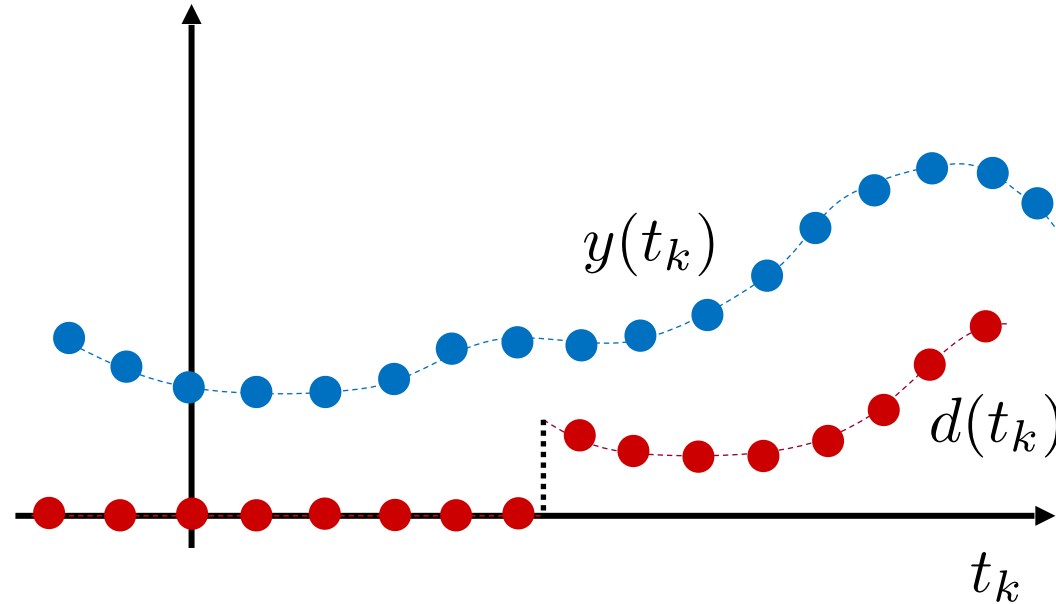




# Digital Implementation of Continuous-Time Feedback Control Systems

## Discrete-time variables

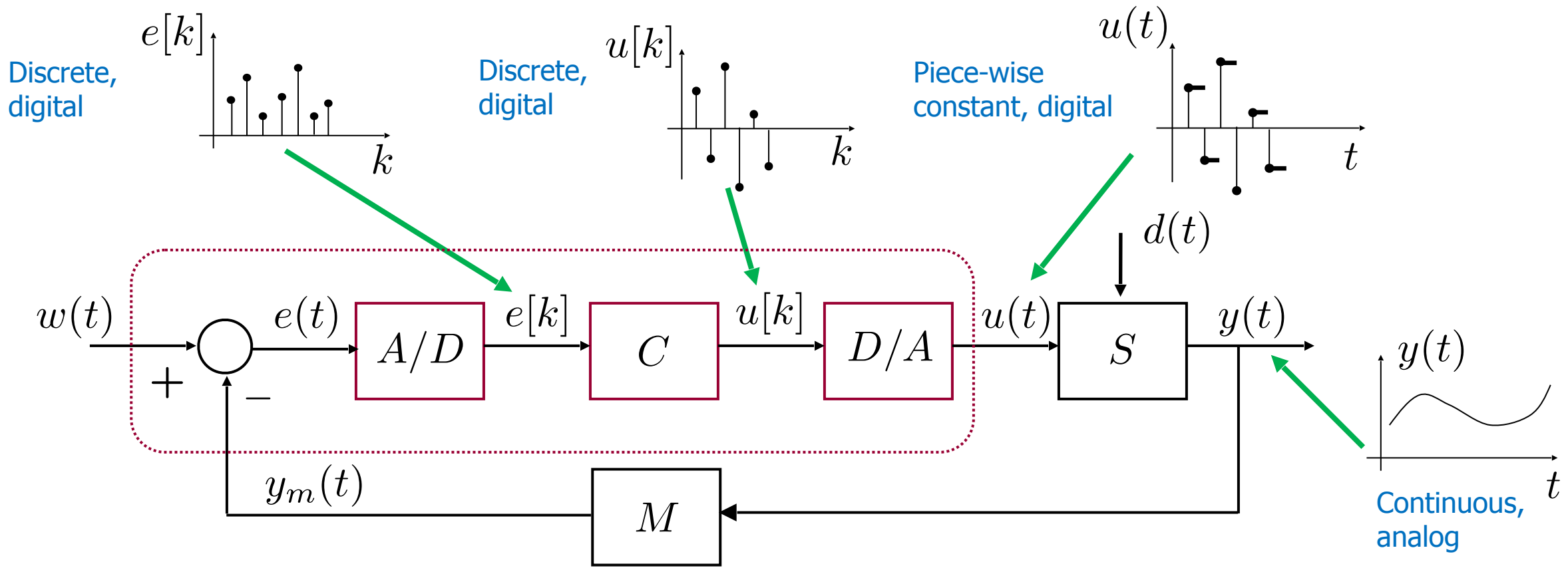
$$\begin{aligned} t_k, k &\in \mathbb{Z} \\ y(t_k) &\in \mathbb{R} \\ d(t_k) &\in \mathbb{R} \\ &\vdots \end{aligned}$$



In case of regular sampling:  $y(t_k) : t_k = kT_s, k \in \mathbb{Z}$

➡  $y[k], d[k], \dots k \in \mathbb{Z}$

# Digital Control: Typical Structure

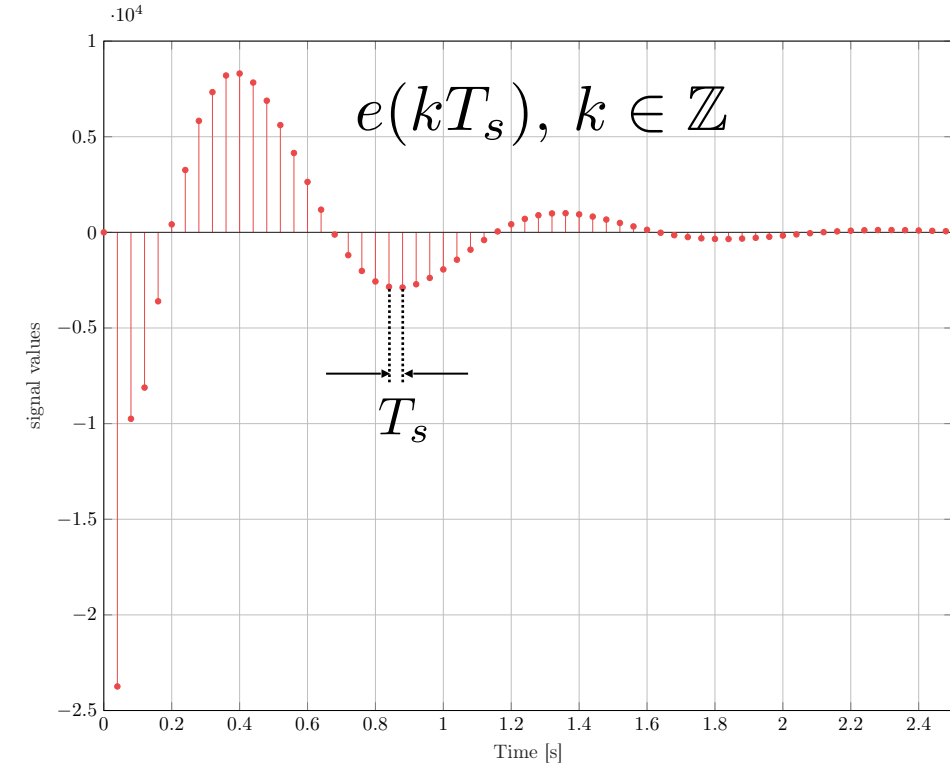
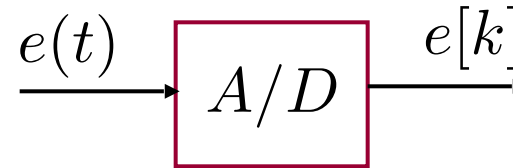
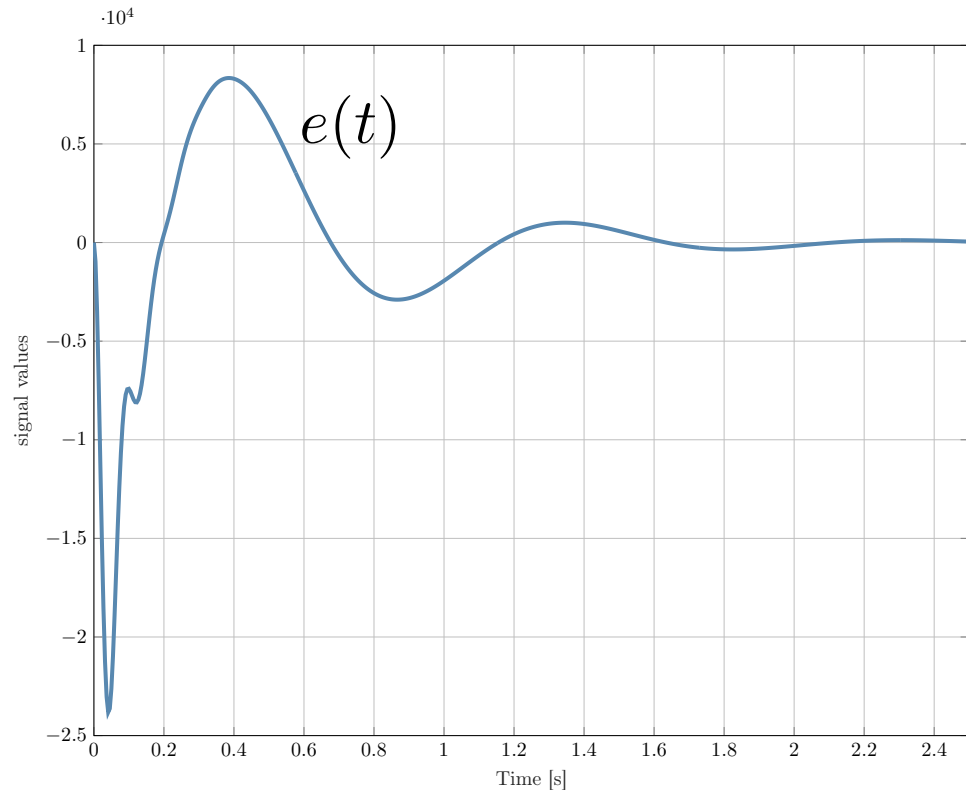


From a mathematical point of view, digital control systems are **hybrid systems** since continuous-time and discrete-time variables are present at the same time

- **Analog** variables: their values change with continuity
- **Digital** variables: their values are quantised

# Analog Controller vs. Digital Controller

- The analog controller receives analog inputs in continuous-time and yields analog outputs in continuous-time
- The digital controller is implemented on a digital computing device (micro-controller, DSP card, PLC, etc.)
- Digital computing devices can only elaborate digital sampled-time variables, and suitable conversion devices are needed at the interface with the controlled system: Analog/Digital Converters (**A/D**) Digital/Analog Converters (**D/A**)
- A/D and D/A converters need to be synchronised via a clock signal with period  $T_s$  (**sampling time**)
- The control unit gets the input variables from the A/D converter and yields the output variable to the D/A converter **only on the clock time-instants**
- Such variables are denoted as **discrete-time** variables
- When there is no risk of ambiguity, we drop the sampling-time notation and only keep the discrete-time index:  $t_k = kT_s \implies k$



**Ideally**, the A/D conversion implements the **sampling mechanism**

$$\{e(t) : t = kT_s, k \in \mathbb{Z}\} \quad \text{green arrow} \quad e[k]$$

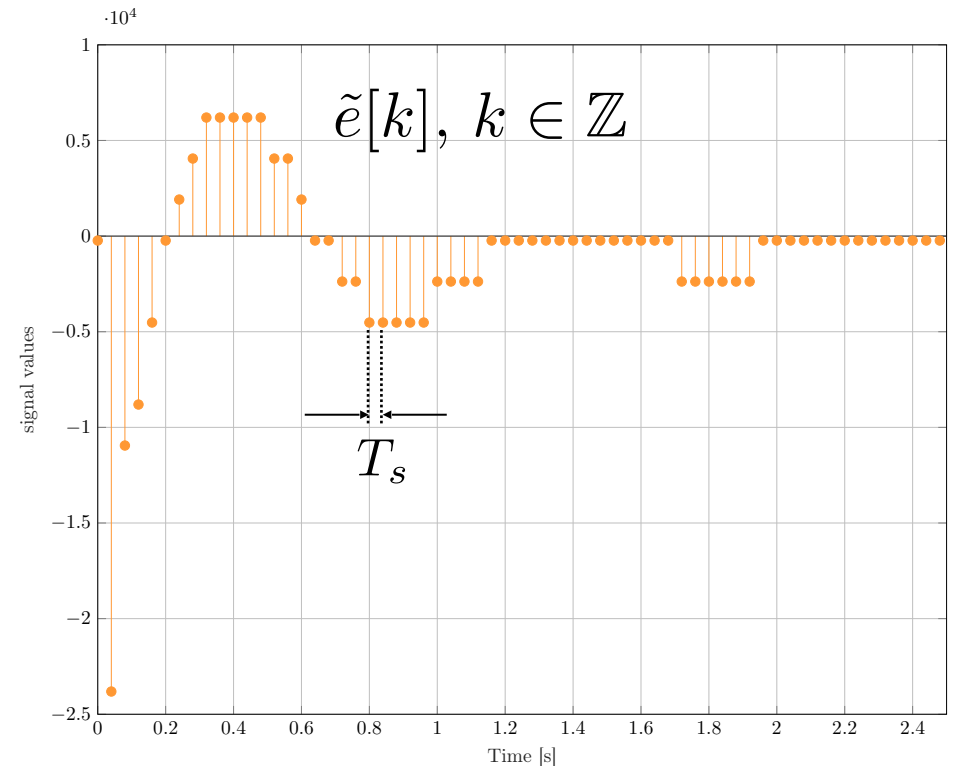
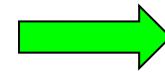
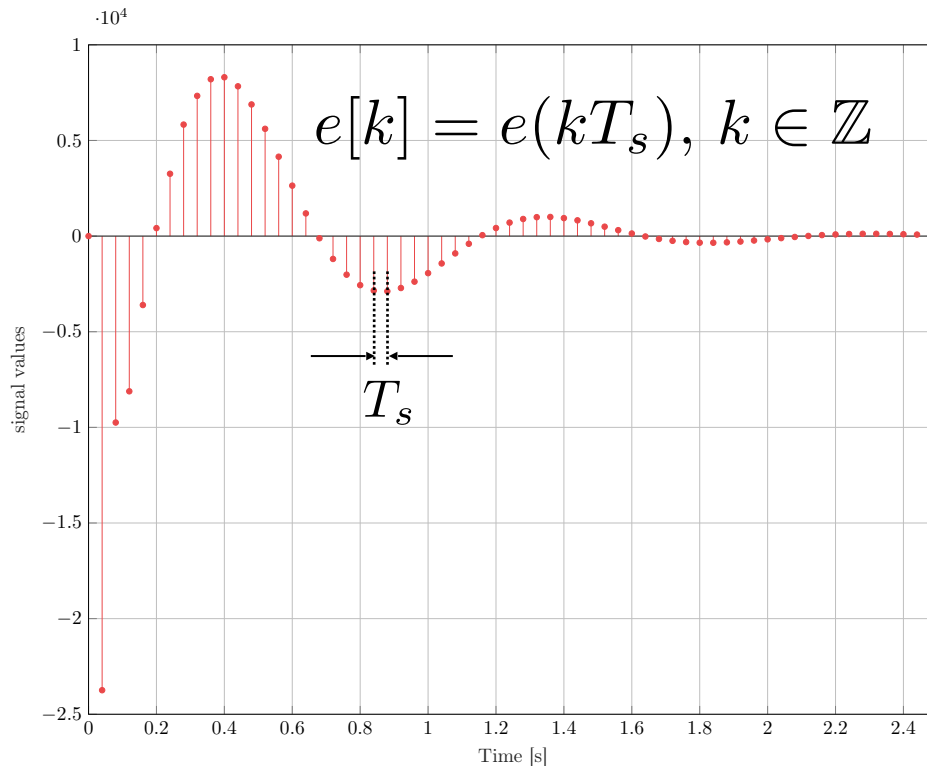
with **sampling frequency**

$$f_s = \frac{1}{T_s}$$

The A/D conversion implies:

- loss of information (continuous-time/discrete-time)
- quantisation noise and distortion (analog to digital)

$$e(t) \xrightarrow{\text{green arrow}} e[k], k \in \mathbb{Z} \xrightarrow{\text{green arrow}} \tilde{e}[k] \neq e[k], k \in \mathbb{Z} \quad 011010\dots$$

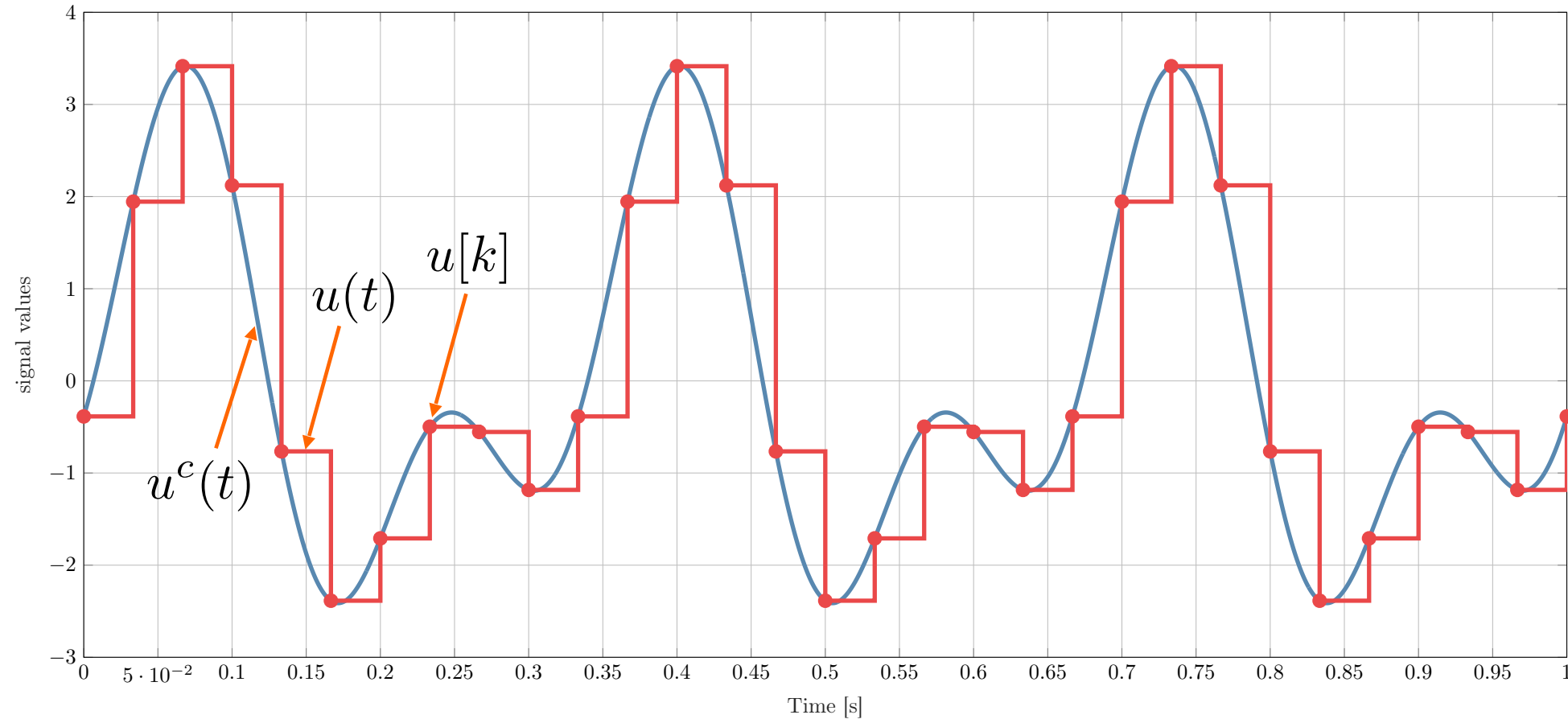


[In this course, we suppose  $\tilde{e}[k] = e[k], k \in \mathbb{Z}$  ]

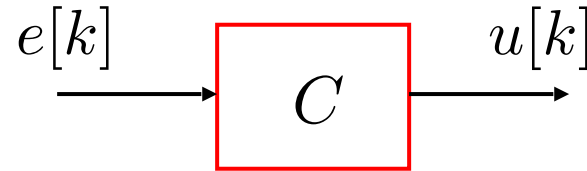


Zero-order hold (ZOH):  $u(t) = u[k], kT_s \leq t < (k+1)T_s, k \in \mathbb{Z}$

The D/A conversion using a ZOH is a stair-wise **delayed** approximation of the unknown underlying continuous time function  $u^c(t)$

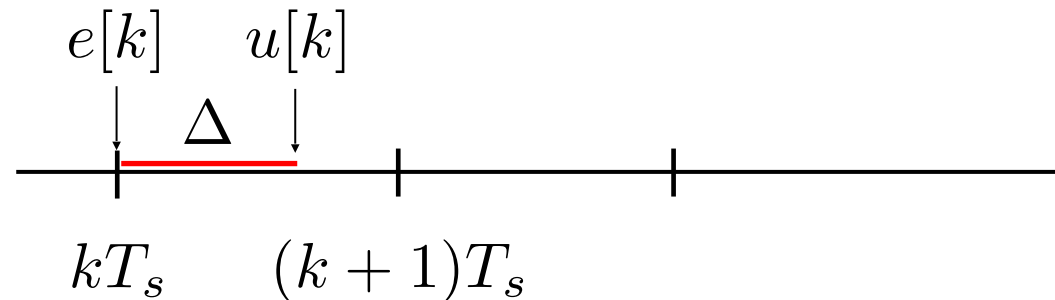


The controller is a discrete-time system, that is: a **computational algorithm!**



$$u[k] = f(u[k-1], u[k-2], \dots, e[k], e[k-1], \dots)$$

**Temporisation:**



The controller computation time should satisfy:  $\Delta < T_s$



- The  $\mathcal{Z}$  transform of a sequence  $\{x[k], k = 0, 1, \dots\}$  with  $x[k] = 0, \forall k < 0$  is defined as:

$$X(z) = \mathcal{Z}\{x[k]\} = \sum_{k=0}^{\infty} x[k]z^{-k}$$

- The  $\mathcal{Z}$  transform, analogously to the Laplace transform, maps discrete-time sequences into functions of the complex variable  $z$
- The  $\mathcal{Z}$  transform is very useful to represent in a compact form difference equations like the ones representing a **control algorithm** of the form:

$$u[k] = f(u[k-1], u[k-2], \dots, e[k], e[k-1], \dots) \quad (\star)$$

- In our introductory course, the  $\mathcal{Z}$  transform will only be used:
  - ✦ **(a)** to obtain in compact form the digital implementation of a continuous-time controller and afterwards
  - ✦ **(b)** to obtain the **control algorithm** of the form  $(\star)$
- The expression of a  $\mathcal{Z}$  transform can be **equivalently** represented in **negative** or **positive** powers of  $z$  as in the following example (the expression in negative powers of  $z$  can be immediately obtained by dividing numerator and denominator by the highest power of  $z$ ):

$$X(z) = \frac{1 - 2z^{-1}}{4 + 6z^{-1} + 8z^{-2}} \iff X(z) = \frac{z^2 - 2z}{4z^2 + 6z + 8}$$

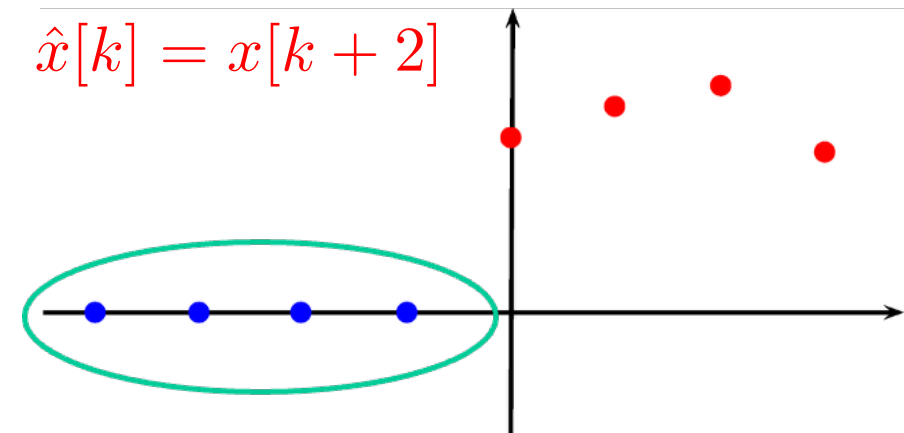
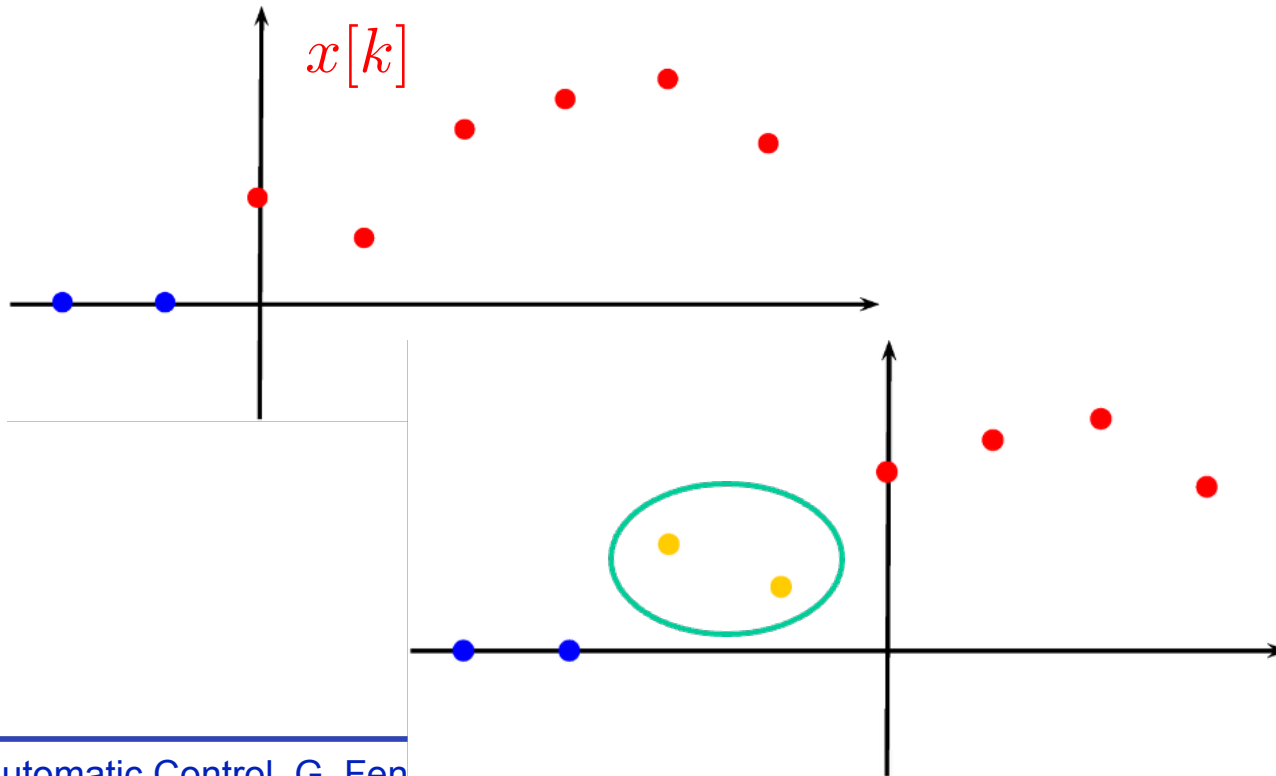
- To obtain the control algorithm, the expression in negative powers of  $z$  will be used.

# Main properties of the Z Transform

**Linearity:**  $\mathcal{Z}\{c_1 f[k] + c_2 g[k]\} = c_1 \mathcal{Z}\{f[k]\} + c_2 \mathcal{Z}\{g[k]\}$

**One-step forward time-translation:**  $\mathcal{Z}\{x[k + 1]\} = z\{X(z) - x[0]\}$

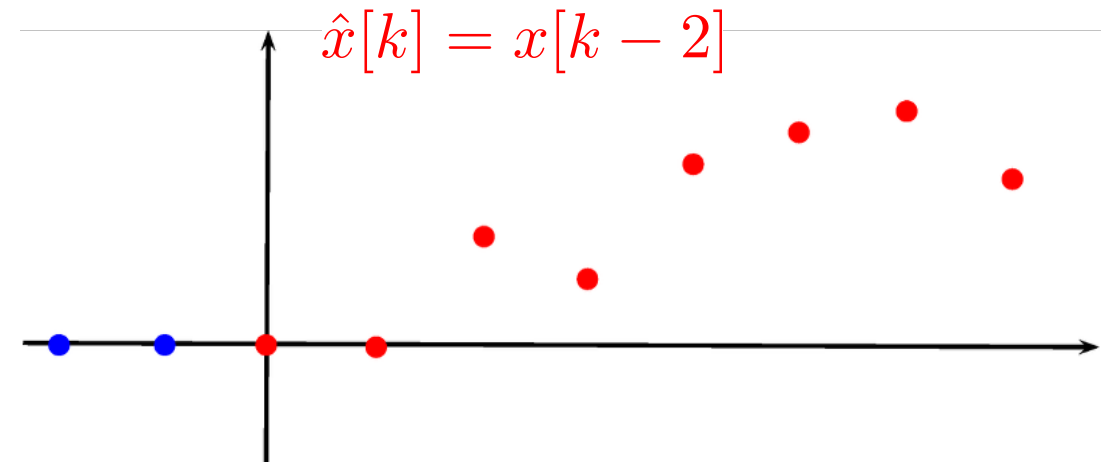
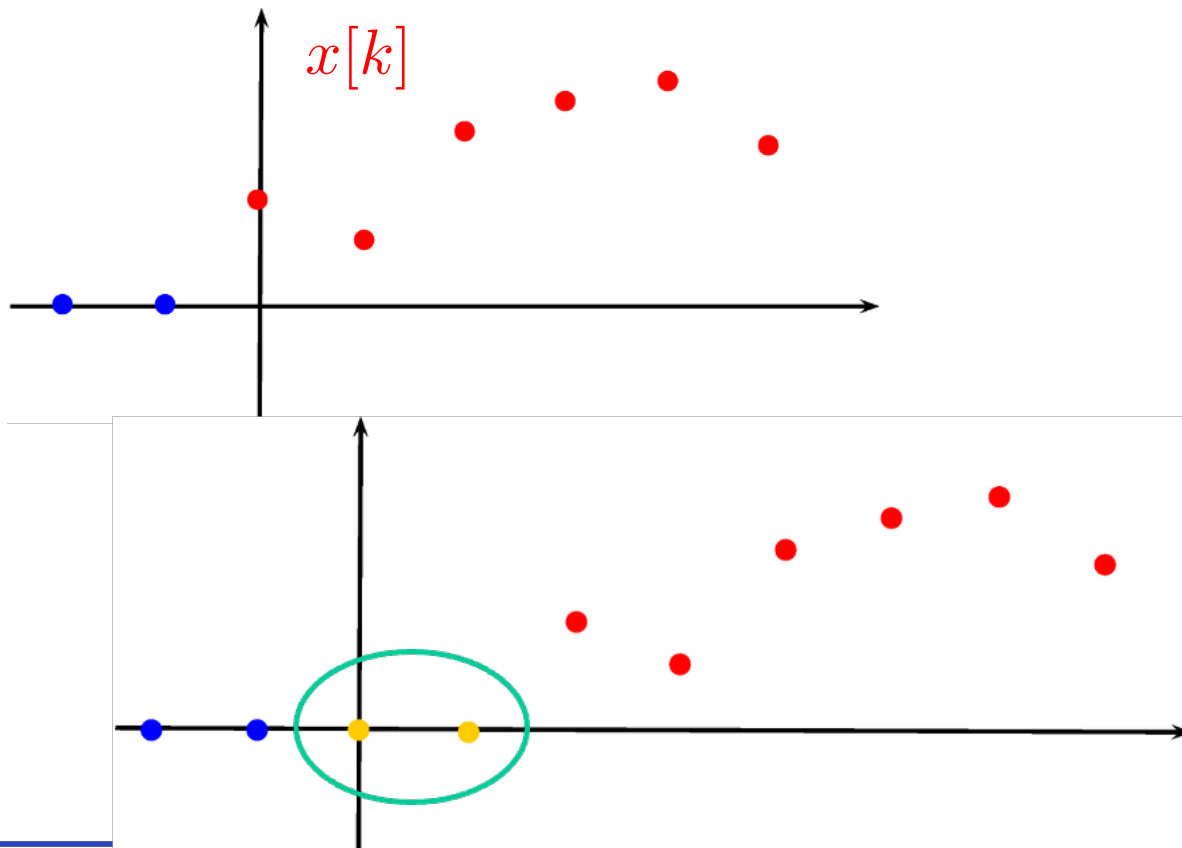
**m-steps forward time-translation:**  $\mathcal{Z}\{x[k + m]\} = z^m \left\{ X(z) - \sum_{k=0}^{m-1} x[k] z^{-k} \right\}$



# Main properties of the Z Transform

**One-step backward time-translation (delay):**  $\mathcal{Z}\{x[k - 1]\} = z^{-1}X(z)$

**m-steps backward time-translation:**  $\mathcal{Z}\{x[k - m]\} = z^{-m}X(z)$



From basic sampling theory:

- If the continuous-time signal  $e(t)$  is **band-limited** with bandwidth  $B = [0, \bar{\omega}]$ , the sampling angular frequency has to satisfy:

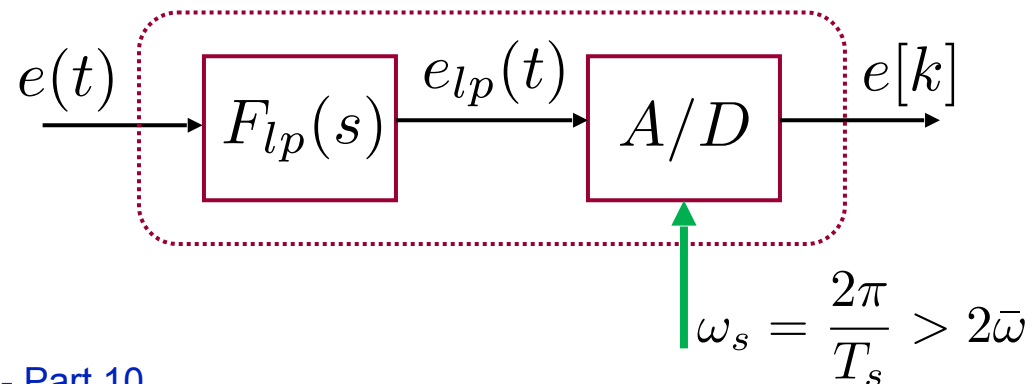
$$\omega_s = \frac{2\pi}{T_s} > 2\bar{\omega}$$

according to the *Nyquist Sampling Criterion*

[Livescripts in MS Teams](#): see Part 10:  
[aliasing\\_demo](#)

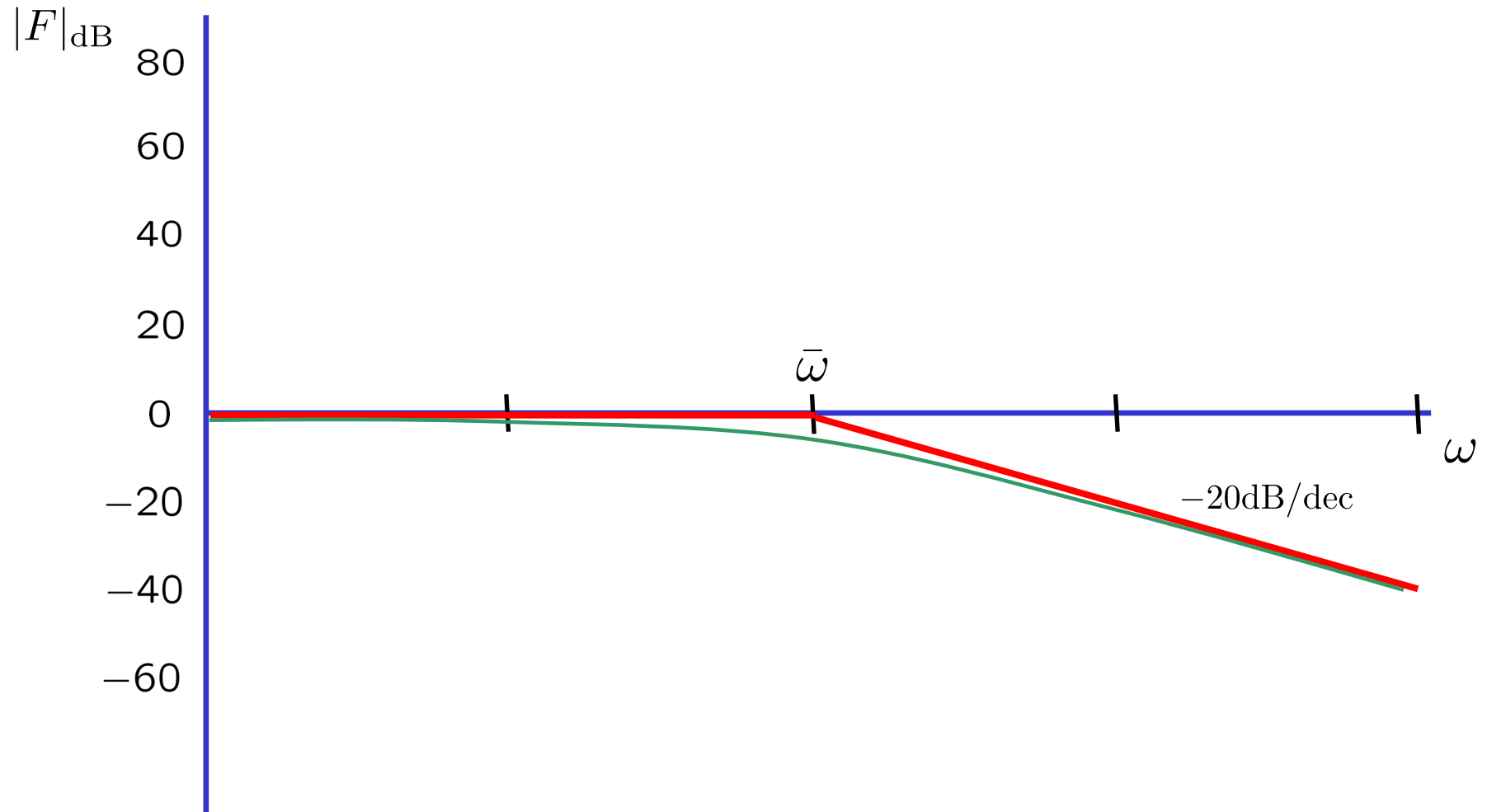


- Since, in general,  $e(t)$  is **not band-limited**, to mitigate the *aliasing phenomenon* an **anti-aliasing lowpass filter**  $F_{lp}(s)$  is designed and used:



For example, a simple first-order anti-aliasing filter could be used:

$$F(s) = \frac{1}{1 + s/\bar{\omega}}$$



# Choice of Sampling Frequency in Digital Control Systems



Recall the frequency response behaviour of the complementary sensitivity function from Part 9, slide 40:

$$|F(j\omega)| = \frac{|L(j\omega)|}{|1 + L(j\omega)|} \simeq \begin{cases} 1 & \text{if } |L(j\omega)| \gg 1 \\ |L(j\omega)| & \text{if } |L(j\omega)| \ll 1 \end{cases}$$

- $F(s)$  **low-pass** filter with bandwidth  $B \simeq [0, \omega_c]$

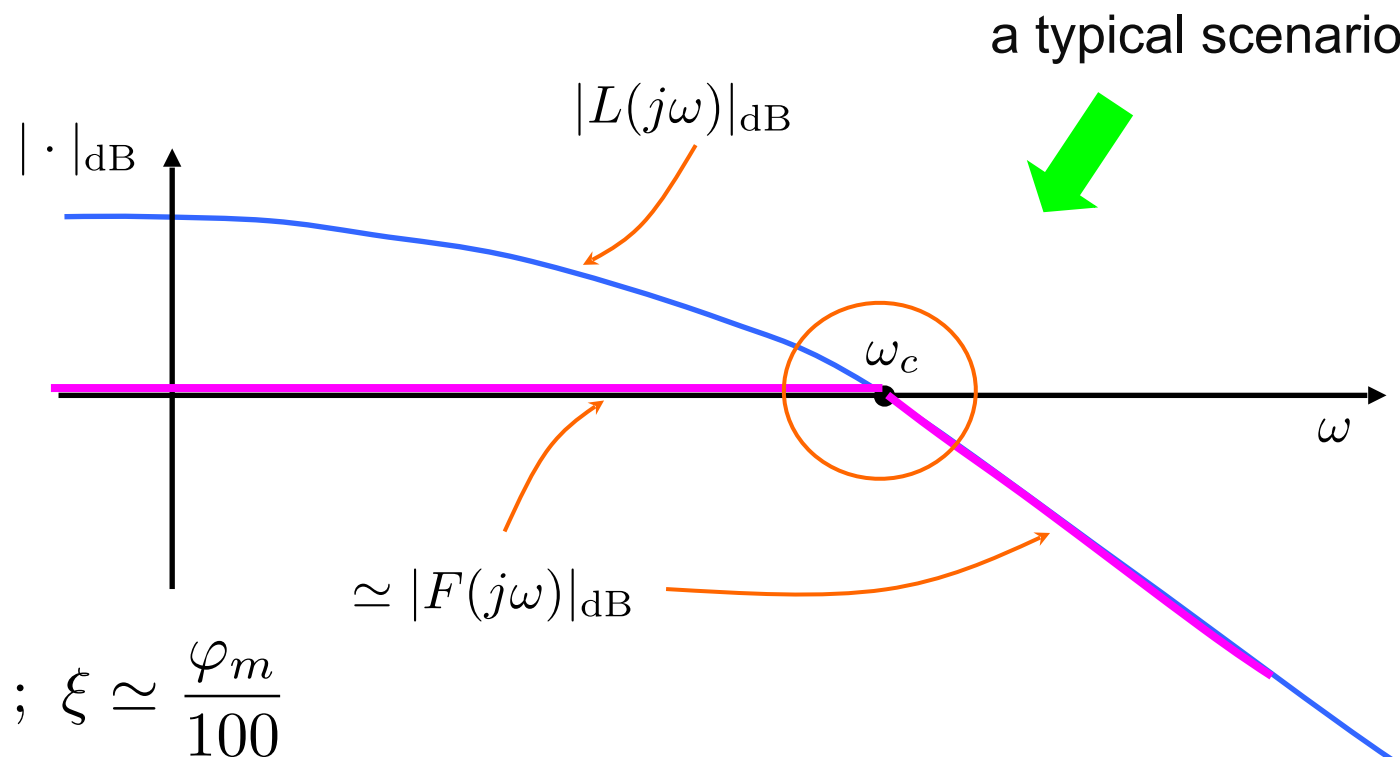
- Gain:

$$\mu_F = \begin{cases} 1 & \text{if } g > 0 \\ \frac{\mu}{1 + \mu} & \text{if } g = 0 \end{cases}$$

- Dominant poles:

– if real:   $\tau \simeq \frac{1}{\omega_c}$

– if complex:   $\omega_n \simeq \omega_c$ ;  $\xi \simeq \frac{\varphi_m}{100}$



(a) Choice of  $\omega_s$  based on the **closed-loop bandwidth**:

- The **closed-loop bandwidth** approximately is  $B_{cl} = [0, \omega_c]$
- **Empirical rule**:

$$\omega_s : \alpha\omega_c < \omega_s < 10\alpha\omega_c \text{ with } \alpha \in [5, 10] \longrightarrow \frac{2\pi}{10\alpha\omega_c} < T_s < \frac{2\pi}{\alpha\omega_c}$$

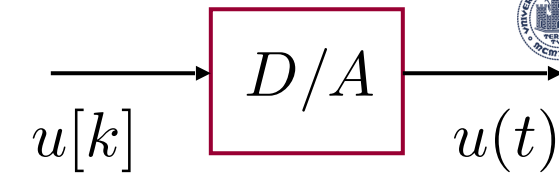
(b) Choice of  $T_s$  based on **number of samples in the transient**:

- The closed-loop system has **one** (with  $\tau \simeq \frac{1}{\omega_c}$ ,  $t_{s,0.01} \simeq 4.6\tau$ ) or **two dominant poles** (with  $\omega_n \simeq \omega_c$ ;  $\xi \simeq \frac{\varphi_m}{100}$ ,  $t_{s,0.01} \simeq \frac{4.6}{\xi\omega_c}$ )
- **Empirical rule**:

$$T_s : \frac{t_{s,0.01}}{10\alpha} < T_s < \frac{t_{s,0.01}}{\alpha} \text{ with } \alpha \in [5, 10]$$

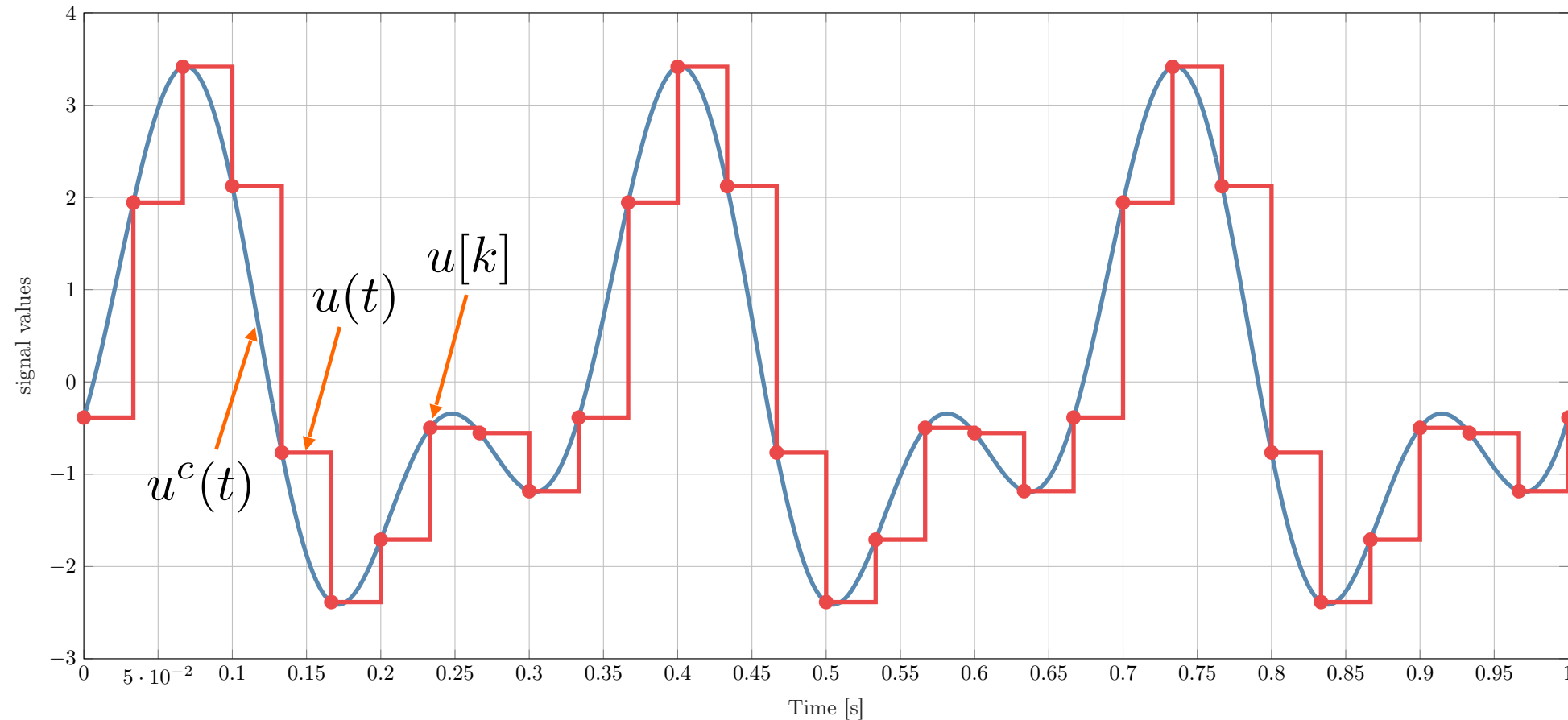


Recall the **D/A conversion** from slide 51:



Zero-order hold (ZOH):  $u(t) = u[k], kT_s \leq t < (k+1)T_s, k \in \mathbb{Z}$

The D/A conversion using a ZOH is a stair-wise **delayed** approximation of the unknown underlying continuous time function  $u^c(t)$



**The ZOH block introduces a time-delay**



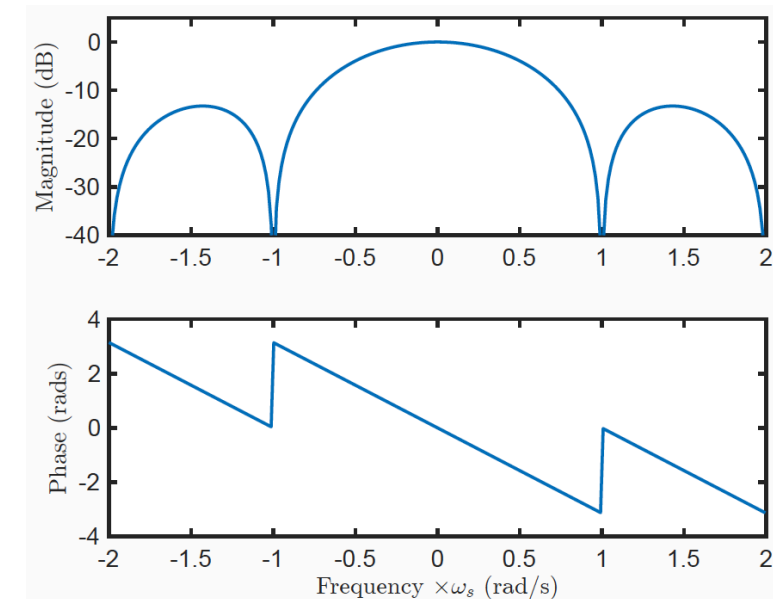
- The ZOH has a frequency response:

$$H_0(j\omega) = 2e^{-j\omega T_s/2} \frac{\sin(\omega T_s/2)}{\omega}$$

The diagram of  $H_0(j\omega)$  is shown on the left for  $T_s = 1\text{sec}$

- The slope of the phase is  $-T_s/2$  which corresponds to a **delay** of  $T_s/2$
- Moreover:

$$|H_0(j\omega)| \simeq 0\text{dB}, \text{ for } \omega \in [0, \bar{\omega}] \text{ with } \bar{\omega} \ll \frac{\omega_s}{2}$$



- From theory of Laplace and Z transforms under ideal sampling it follows that

$$\begin{array}{ccc} z = e^{sT_s} & & s = \frac{1}{T_s} \ln z \\ G(z) \xrightarrow{\quad} G(s) & & G(s) \xrightarrow{\quad} G(z) \end{array}$$

- If  $G(s)$  and  $G(z)$  are **transfer functions**, these relations **should not be directly used** (for example, direct substitution would not lead to ratios of polynomials!)
- Moreover, it is of key importance to ensure that the approximate transformations from  $s$  to  $z$  and viceversa **preserve the stability properties**
- In the context of constructing a digital implementation of a continuous-time feedback control system, we are interested in the approximate transformation from  $s$  to  $z$
- In the course, we consider the **Backward Euler** and the **Tustin** approximations

- The **Backward Euler** (BE) approximate transformation:

$$s = \frac{z - 1}{T_s z}$$
$$G(s) \quad \longrightarrow \quad \hat{G}_{\text{BE}}(z)$$

The BE approximation **preserves the stability properties** (asymptotically stable poles of  $G(s)$  are mapped into asymptotically stable poles of  $\hat{G}_{\text{BE}}(z)$  )

- The **Tustin** (Tu) approximate transformation:

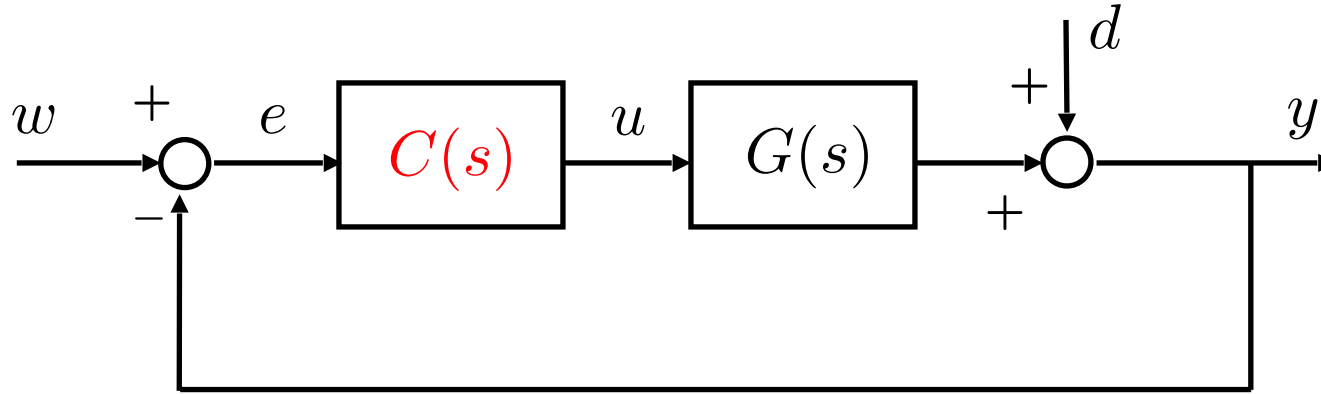
$$s = \frac{2}{T_s} \frac{z - 1}{z + 1}$$
$$G(s) \quad \longrightarrow \quad \hat{G}_{\text{Tu}}(z)$$

The Tu approximation **preserves the stability properties** (asymptotically stable poles of  $G(s)$  are mapped into asymptotically stable poles of  $\hat{G}_{\text{Tu}}(z)$  )

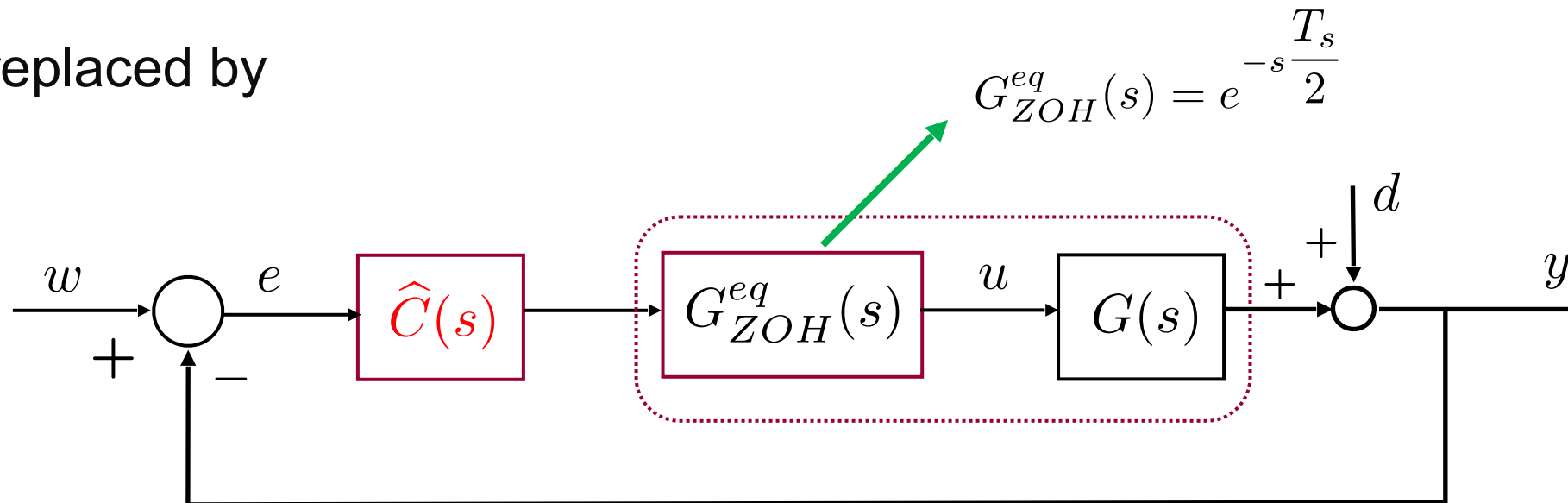
# Digital Implementation of C.T. Feedback Control Systems

The following **operational procedure** is carried out:

(1) the original control system



is replaced by



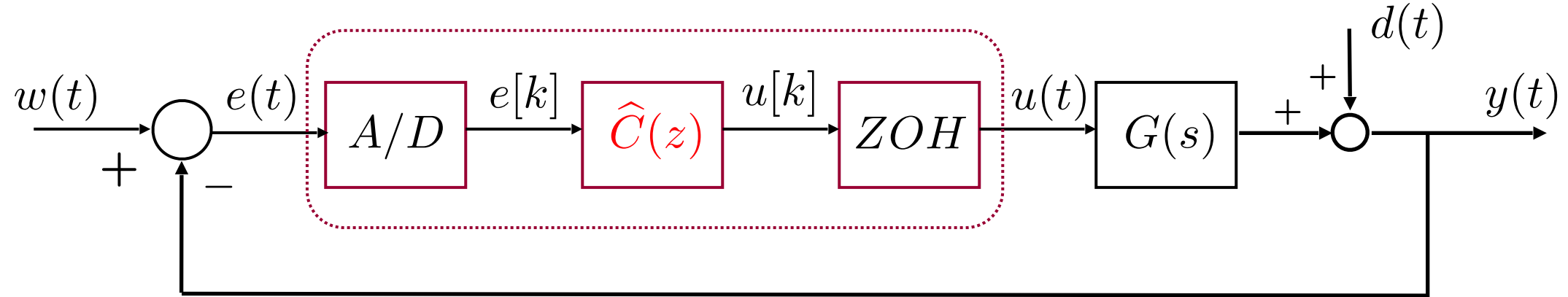
**Remark:** The addition of the block  $G_{ZOH}^{eq}(s)$  takes into account the **delay introduced by the ZOH**. This additional block does not influence the requirements on closed-loop bandwidth and gain margin.

- (2) the controller  $\hat{C}(s)$  is designed in continuous-time by loop-shaping techniques ensuring that the original specifications are met
- (3) the sampling-time  $T_s$  is selected according to the empirical rules in [Slide 60](#)
- (4) the discrete-time approximation  $\hat{C}(z)$  of the continuous-time controller  $\hat{C}(s)$  is computed by the **BE** or by the **Tu** transformation:

$$s = \frac{z - 1}{T_s z}$$
$$\hat{C}(s) \xrightarrow{\text{BE}} \hat{C}_{\text{BE}}(z)$$

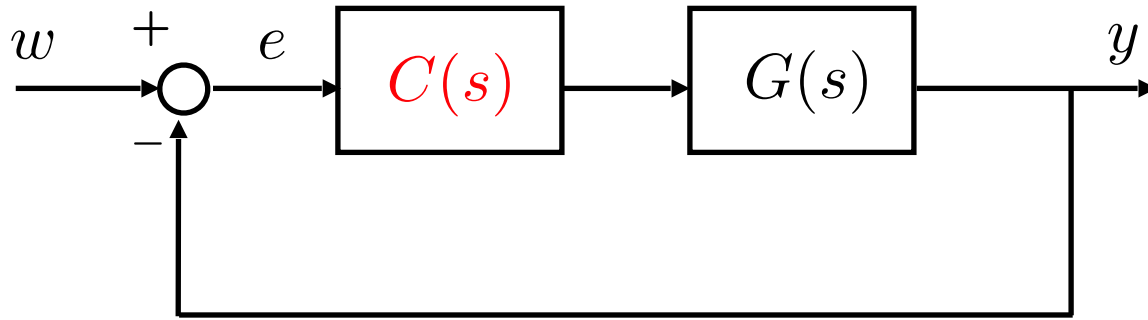
$$s = \frac{2}{T_s} \frac{z - 1}{z + 1}$$
$$\hat{C}(s) \xrightarrow{\text{Tu}} \hat{C}_{\text{Tu}}(z)$$

(5) the **digital control system** is implemented:



(6) the digital implementation is constructed via numerous **approximations**. Hence, the performance of the digitally-controlled system has to be verified *a posteriori* with the possible need for an improved re-design

## Example - Digital Implementation of C.T. Control Systems



$$G(s) = 0.1 \cdot \frac{(1 - 2s)}{s(1 + 10s)(1 + 0.1s)}$$

**Design specifications (in continuous-time):**  $\varphi_m \geq 60^\circ$

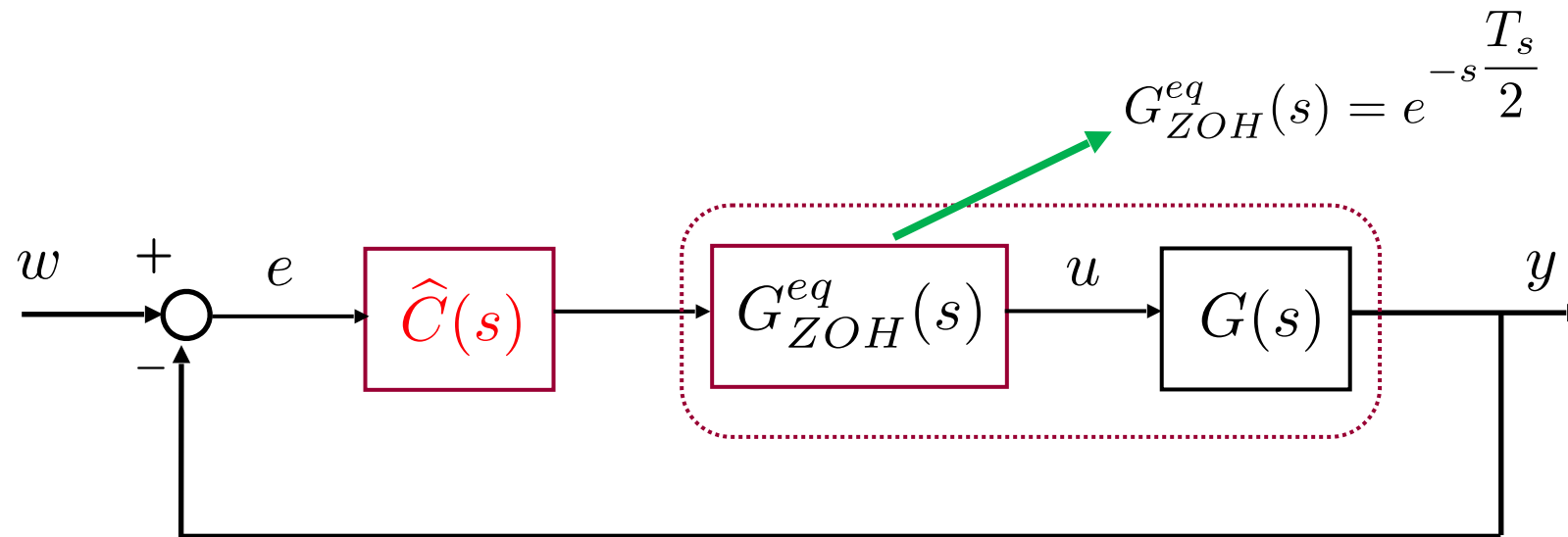
- The controlled system has the same pole-zero configuration of the non-minimum phase system considered in [Slide 35](#) and we choose the controller considered in Attempt 2:

$$C(s) = \mu_C \frac{1 + 10s}{1 + 0.1s} \quad \longrightarrow \quad L(s) = \mu_C \frac{0.1 \cdot (1 - 2s)}{s(1 + 0.1s)^2}$$



## Example - Digital Implementation of C.T. Control System (contd.)

- The **design** of the controller needs to take into account suitable **additional margins** so that the original specifications are met with the modified scheme below that takes into account the **digital implementation**:



- Hence, we design:

$$\hat{C}(s) = \hat{\mu}_C \frac{1 + 10s}{1 + 0.1s} \quad \text{such that} \quad \varphi_m \geq 60^\circ + \delta_\varphi$$

where  $\delta_\varphi$  is sufficiently large so that the ZOH-induced delay can be accommodated

# Example - Digital Implementation of C.T. Control System (contd.)

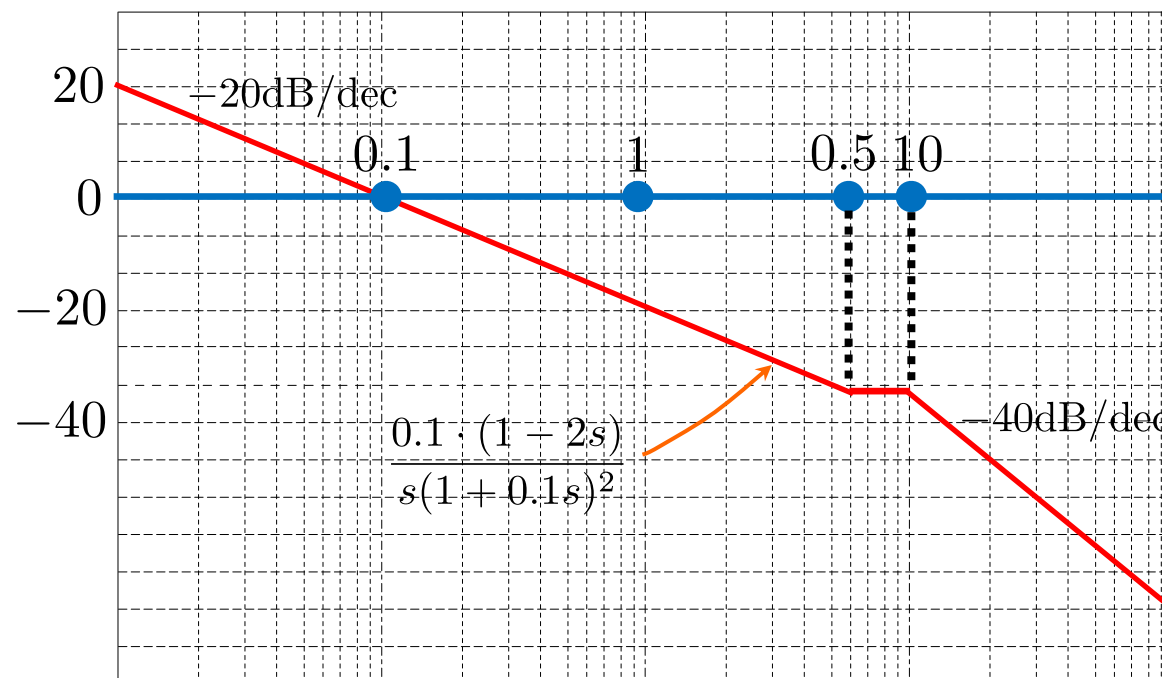
- We set:  $\delta_\varphi \geq 5^\circ$

- Consider:

$$\hat{L}(s) = \hat{\mu}_C \frac{0.1 \cdot (1 - 2s)}{s(1 + 0.1s)^2}$$

- For  $\hat{\mu}_C = 2$  let's compute

$$\omega_c : |\hat{L}(j\omega_c)|_{\text{dB}} = 0$$



$$\downarrow -20\log\omega + 20\log(2 \cdot 0.1) = 0 \quad \Rightarrow \quad \omega_c \simeq 0.2 \text{ rad/sec}$$

$$\downarrow \varphi_c = \arg L(j\omega_c) = -90^\circ - \arctan(2 \cdot 0.2) \cdot \frac{180^\circ}{\pi} + \arg \frac{1}{(1 + 0.1 \cdot 0.2j)^2}$$

$$\simeq -114.3^\circ \quad \Rightarrow \quad \varphi_m \simeq 66^\circ \quad \Rightarrow \quad \delta_\varphi \simeq 6^\circ$$

## Example - Digital Implementation of C.T. Control System (contd.)

- Recall the empirical rule on Slide 60:

$$\omega_s : \alpha\omega_c < \omega_s < 10\alpha\omega_c \text{ with } \alpha \in [5, 10] \longrightarrow \frac{2\pi}{10\alpha\omega_c} < T_s < \frac{2\pi}{\alpha\omega_c}$$

- Since  $\omega_c \simeq 0.2 \text{ rad/sec}$

$$\longrightarrow \frac{\pi}{\alpha} < T_s < \frac{10\pi}{\alpha} \text{ with } \alpha \in [5, 10] \longrightarrow \text{a suitable choice is } T_s = 1 \text{ sec}$$

- Now:  $\omega_s = \frac{2\pi}{T_s} = 2\pi \simeq 6.28 \text{ rad/sec} \longrightarrow \omega_c \simeq 0.2 \text{ rad/sec} \ll \frac{\omega_s}{2} \simeq 3.14 \text{ rad/sec}$

- Hence (recall Slide 65) the effect of the ZOH can be **approximated** by

$$G_{ZOH}^{eq}(s) = e^{-s\frac{T_s}{2}} = e^{-\frac{s}{2}}$$

which introduces a phase contribution of:  $-\frac{\omega_c T_s}{2} \cdot \frac{180^\circ}{\pi} = -\frac{0.2}{2} \cdot \frac{180^\circ}{\pi} \simeq -5.7^\circ$

$\longrightarrow$  the phase margin becomes  $\hat{\varphi}_m = \varphi_m - 5.7^\circ \simeq 60^\circ$

# Example - Digital Implementation of C.T. Control System (contd.)

- Recall the approximate transformations for the digital controllers on Slide 66:

$$s = \frac{z - 1}{T_s z}$$

$$\hat{C}(s) \longrightarrow \hat{C}_{BE}(z)$$

$$s = \frac{2}{T_s} \frac{z - 1}{z + 1}$$

$$\hat{C}(s) \longrightarrow \hat{C}_{Tu}(z)$$

- We obtain (recall that  $T_s = 1\text{sec}$ ):

$$\hat{C}_{BE}(z) = 2 \cdot \left. \frac{1 + 10s}{1 + 0.1s} \right|_{s=\frac{z-1}{z}} = \dots = 20 \cdot \frac{z - 10/11}{z - 1/11} = 20 \cdot \frac{1 - 10/11z^{-1}}{1 - 1/11z^{-1}}$$

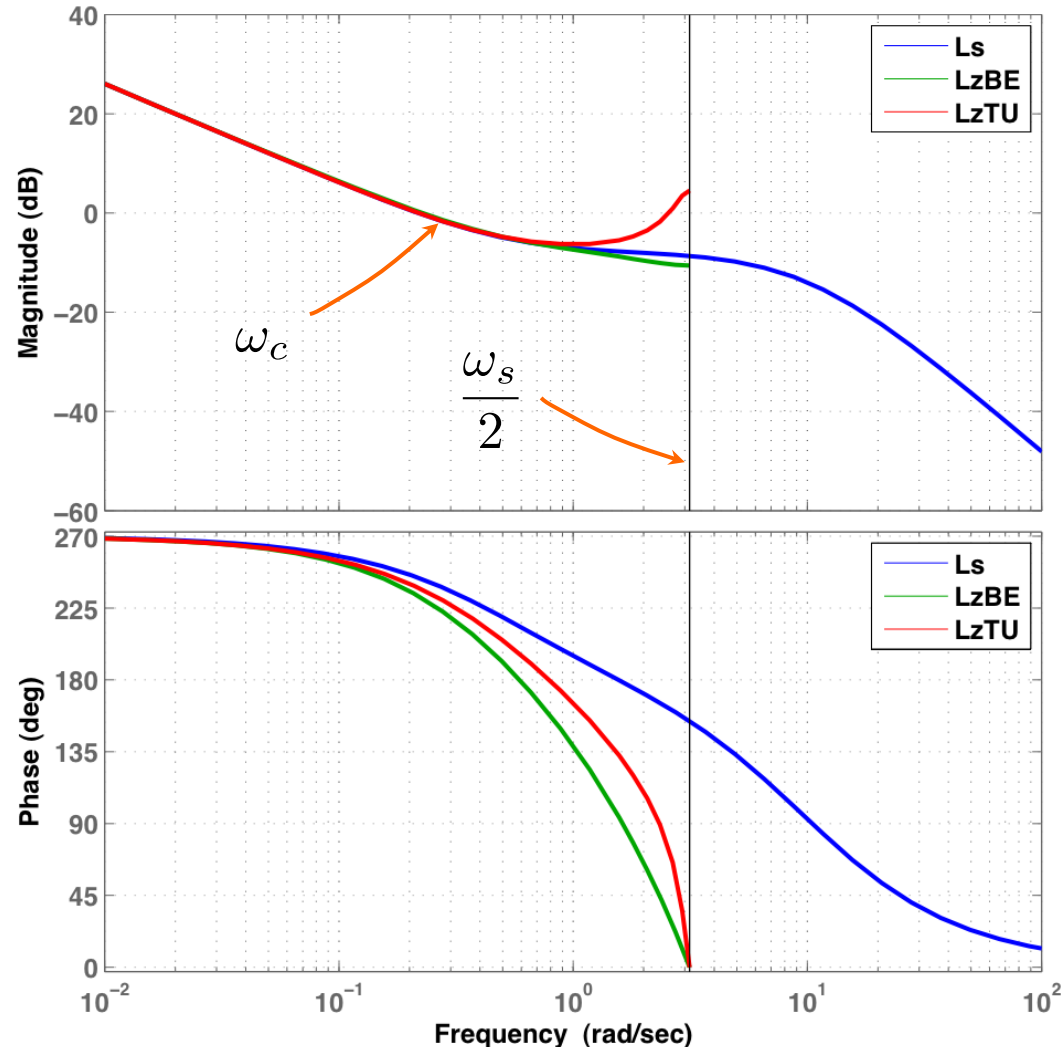
$$\hookrightarrow u[k] = \frac{1}{11}u[k-1] + 20e[k] - \frac{200}{11}e[k-1]$$

$$\hat{C}_{Tu}(z) = 2 \cdot \left. \frac{1 + 10s}{1 + 0.1s} \right|_{s=2\frac{z-1}{z+1}} = \dots = 35 \cdot \frac{z - 19/21}{z + 2/3} = 35 \cdot \frac{1 - 19/21z^{-1}}{1 + 2/3z^{-1}}$$

$$\hookrightarrow u[k] = -\frac{2}{3}u[k-1] + 35e[k] - \frac{95}{3}e[k-1]$$

*A posteriori* performance verification:

Bode Diagram



$$T_s = 1\text{sec}$$

$\hat{C}(s) : K_m \simeq 7.74 \text{ dB at } 1.56 \text{ rad/sec}; \varphi_m \simeq 63.9^\circ \text{ at } 0.218 \text{ rad/sec}$

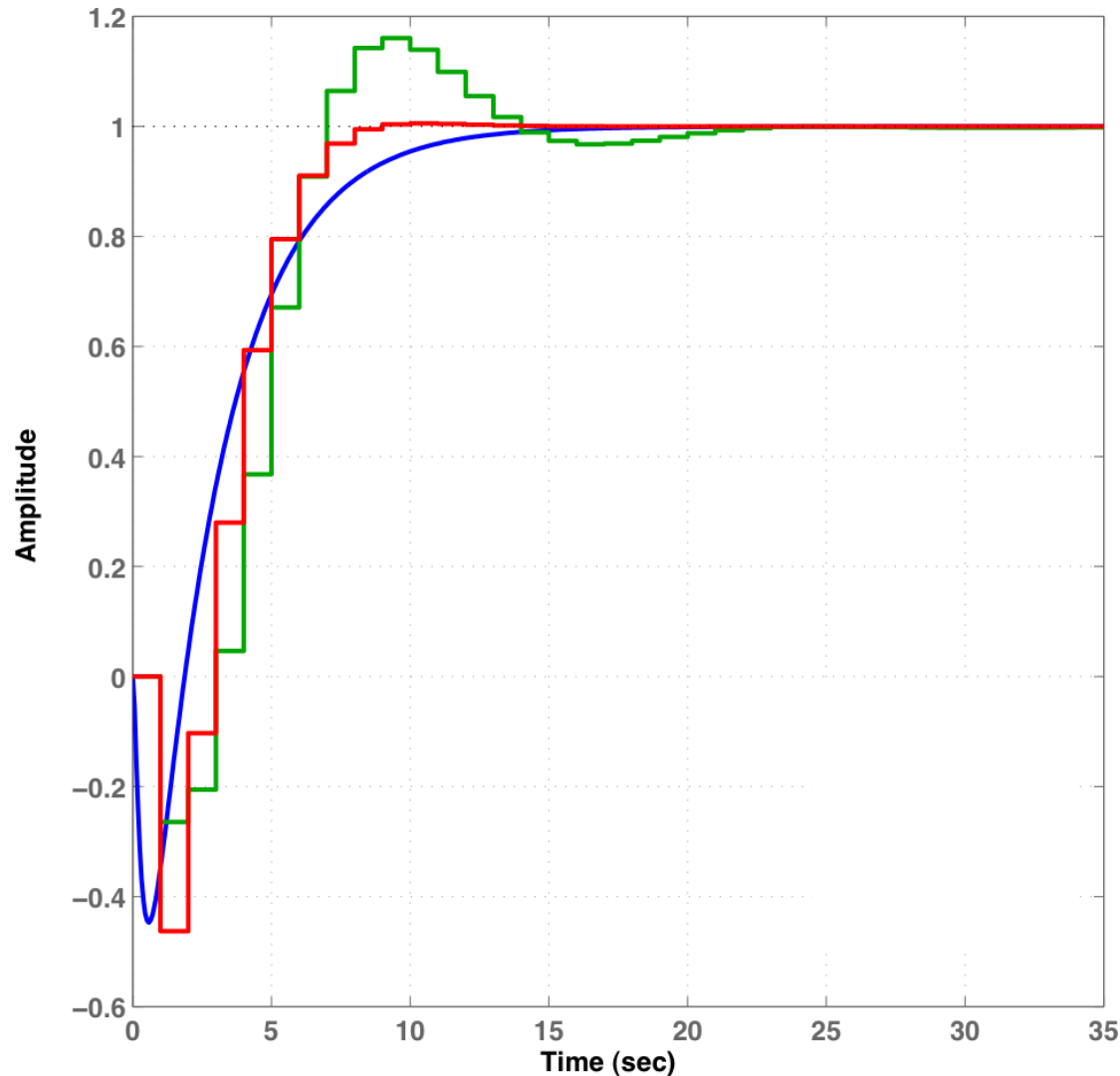
$\hat{C}_{BE}(z) : K_m \simeq 5.56 \text{ dB at } 0.592 \text{ rad/sec}; \varphi_m \simeq 51^\circ \text{ at } 0.227 \text{ rad/sec}$

$\hat{C}_{Tu}(z) : K_m \simeq 6.13 \text{ dB at } 0.791 \text{ rad/sec}; \varphi_m \simeq 57.7^\circ \text{ at } 0.218 \text{ rad/sec}$

**Remark:** as expected, there is a performance deterioration. Depending on the requirements, a re-design might be needed

## Step-response simulation verification:

Step Response



$$T_s = 1\text{sec}$$

$$\hat{C}(s) : K_m \simeq 7.74 \text{ dB at } 1.56 \text{ rad/sec}; \varphi_m \simeq 63.9^\circ \text{ at } 0.218 \text{ rad/sec}$$

$$\hat{C}_{\text{BE}}(z) : K_m \simeq 5.56 \text{ dB at } 0.592 \text{ rad/sec}; \varphi_m \simeq 51^\circ \text{ at } 0.227 \text{ rad/sec}$$

$$\hat{C}_{\text{TU}}(z) : K_m \simeq 6.13 \text{ dB at } 0.791 \text{ rad/sec}; \varphi_m \simeq 57.7^\circ \text{ at } 0.218 \text{ rad/sec}$$

- For comparison, let us now set a **smaller sampling time**:  $T_s = 0.1\text{sec}$
- Now (recall Slide 65) the effect of the ZOH can be approximated by

$$G_{ZOH}^{eq}(s) = e^{-s \frac{T_s}{2}} = e^{-0.05s}$$

which introduces a phase contribution of:  $-\frac{\omega_c T_s}{2} \cdot \frac{180^\circ}{\pi} = -\frac{0.02}{2} \cdot \frac{180^\circ}{\pi} \simeq -0.57^\circ$

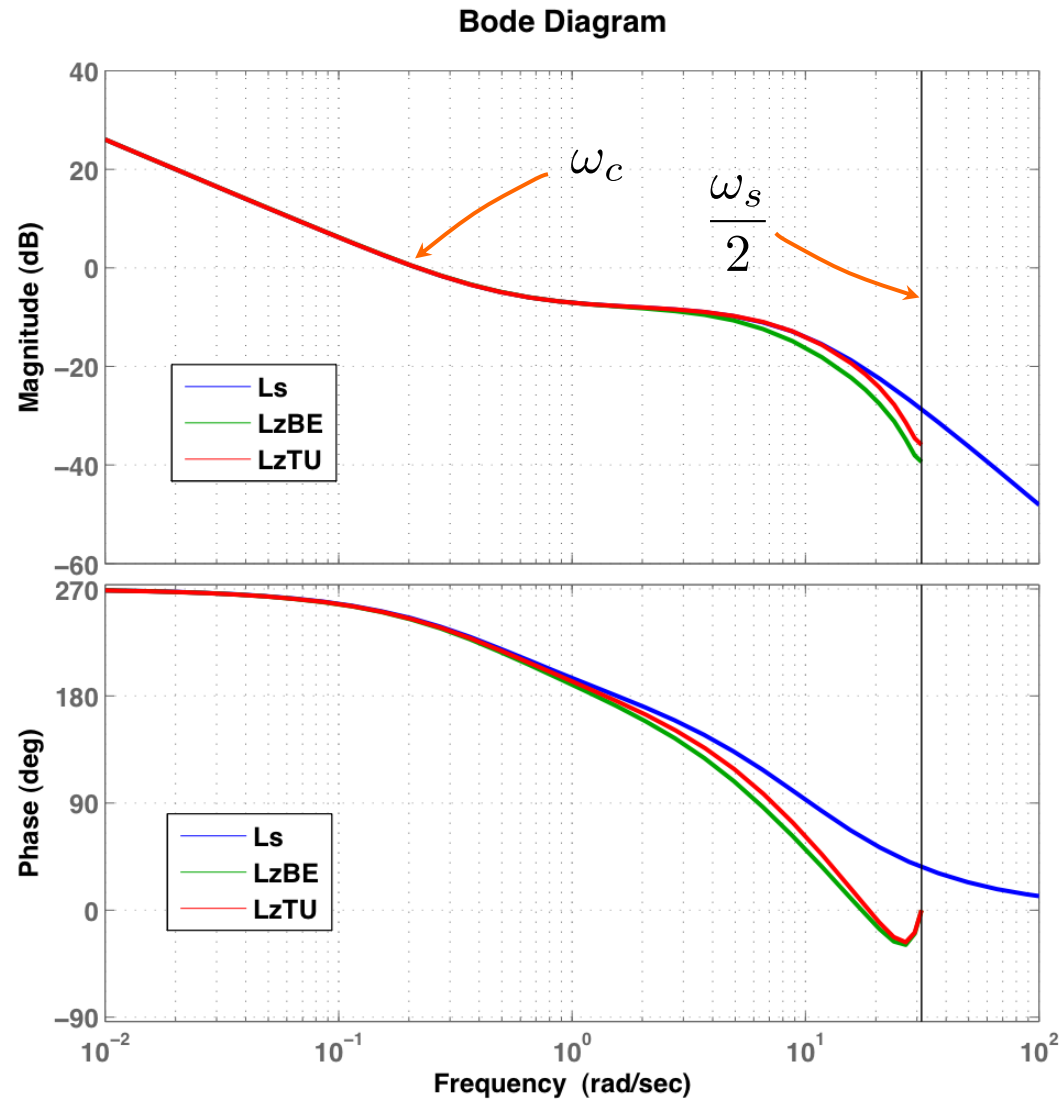
↳ the phase margin becomes  $\hat{\varphi}_m = \varphi_m - 0.57^\circ \simeq 65.4^\circ$

- We obtain:

$$\hat{C}_{BE}(z) = 2 \cdot \frac{1 + 10s}{1 + 0.1s} \bigg|_{s=\frac{z-1}{0.1z}} = \dots = 101 \cdot \frac{z - 100/101}{z - 1/2} = 101 \cdot \frac{1 - 100/101 z^{-1}}{1 - 1/2 z^{-1}}$$

$$\hat{C}_{Tu}(z) = 2 \cdot \frac{1 + 10s}{1 + 0.1s} \bigg|_{s=\frac{2}{0.1} \frac{z-1}{z+1}} = \dots = 134 \cdot \frac{z - 199/201}{z - 1/3} = 134 \cdot \frac{1 - 199/201 z^{-1}}{1 - 1/3 z^{-1}}$$

*A posteriori* performance verification:



$$T_s = 0.1 \text{ sec}$$

$\hat{C}(s)$  :  $K_m \simeq 7.74 \text{ dB}$  at  $1.56 \text{ rad/sec}$ ;  $\varphi_m \simeq 63.9^\circ$  at  $0.218 \text{ rad/sec}$

$\hat{C}_{BE}(z)$  :  $K_m \simeq 7.5 \text{ dB}$  at  $1.27 \text{ rad/sec}$ ;  $\varphi_m \simeq 62.7^\circ$  at  $0.219 \text{ rad/sec}$

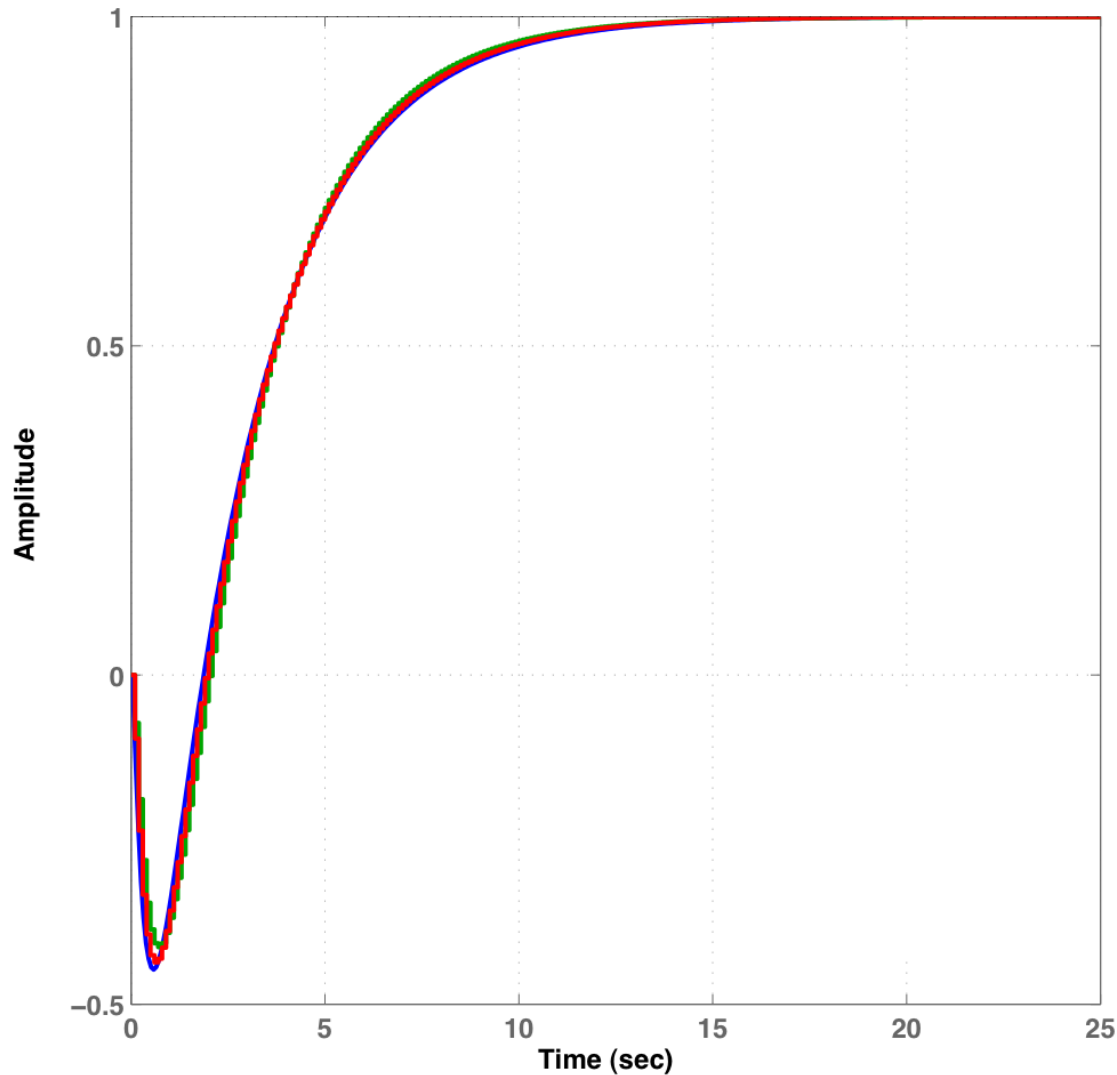
$\hat{C}_{Tu}(z)$  :  $K_m \simeq 7.59 \text{ dB}$  at  $1.39 \text{ rad/sec}$ ;  $\varphi_m \simeq 63.3^\circ$  at  $0.218 \text{ rad/sec}$

**Remark:** as expected, there is **much less performance deterioration** using a smaller sampling-time



## Step-response simulation verification:

Step Response



$$T_s = 0.1\text{sec}$$

$\hat{C}(s) : K_m \simeq 7.74 \text{ dB at } 1.56 \text{ rad/sec}; \varphi_m \simeq 63.9^\circ \text{ at } 0.218 \text{ rad/sec}$

$\hat{C}_{\text{BE}}(z) : K_m \simeq 7.5 \text{ dB at } 1.27 \text{ rad/sec}; \varphi_m \simeq 62.7^\circ \text{ at } 0.219 \text{ rad/sec}$

$\hat{C}_{\text{TU}}(z) : K_m \simeq 7.59 \text{ dB at } 1.39 \text{ rad/sec}; \varphi_m \simeq 63.3^\circ \text{ at } 0.218 \text{ rad/sec}$

# Fundamentals of Automatic Control

... The End ...

